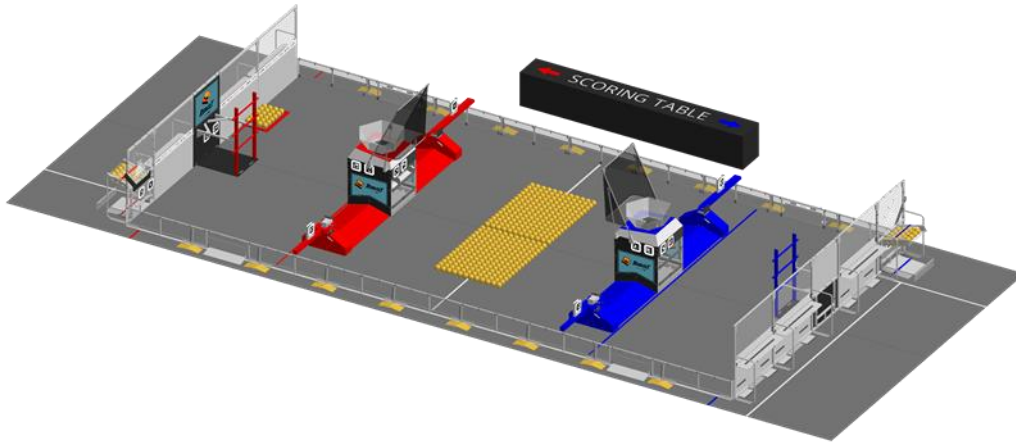


4 Game Overview



In REBUILT™ presented by Haas, two competing alliances are invited to score fuel, cross obstacles, and climb the tower before time runs out. Alliances earn additional rewards for meeting specific scoring thresholds.

During the first 20 seconds of the match, robots are autonomous. Without guidance from their drivers, robots score fuel into their hub. Fuel can be pre-loaded into a robot, obtained from the human player, collected at the depot, or picked up throughout the center of the field. Some robots may also climb the tower to obtain additional points.

During the remaining 2 minutes and 20 seconds, drivers control their robots. Based on the result of autonomous play, alliance hubs will alternate between active and inactive, shifting gameplay between both sides of the field. Robots can collect fuel at any point in the match and may control any amount of fuel at a time. Drivers control their robots to score fuel into their hub while it is active and may perform defensive strategies or collect more fuel while their hub is inactive.

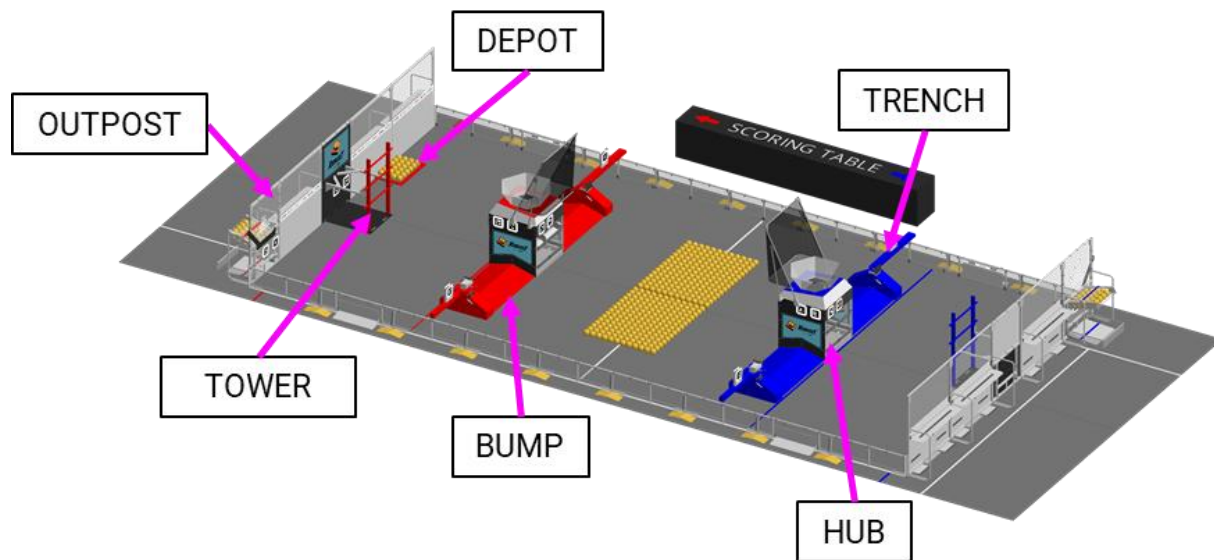
As time runs out, all hubs become active, allowing all robots to score. Robots can climb to the tower's highest heights to score additional points and claim match bonuses to increase their position in the rankings.

The alliance that earns the most points wins the match!

5 ARENA

The ARENA includes all elements of the game infrastructure that are required to play REBUILT™ presented by Haas: the FIELD, SCORING ELEMENTS, queue area, team media area (when available), designated TECHNICIAN area, and all equipment needed for FIELD control, ROBOT control, and scorekeeping.

Figure 5-1 REBUILT ARENA (queue area, TECHNICIAN area, and media area not pictured)



5.1 Dimensions and Accuracy

All official models for the REBUILT FIELD were created in Onshape.



The specification for the REBUILT FIELD can be retrieved from a few locations:

- The 3D CAD model is the official representation of the REBUILT FIELD and how it is constructed.
- Illustrations included in this section are for a general visual understanding of the REBUILT ARENA, and dimensions included in the manual are nominal and no tolerances are implied. Please refer to the official drawings for exact dimensions, tolerances, and construction details.
- The [Field Dimension Drawings](#) package has critical dimensions for each FIELD element.
- The [FIELD Manual](#) includes instructions on how to build the FIELD along with showing the ways construction type will influence the field tolerances. It also includes many of the key dimensions which are listed in the Official FIELD Drawings.
- The FIELD Acceptance Checklists for the [welded FIELD](#) and for the [AndyMark FIELD](#) include the controlled dimensions (with relevant tolerances) which will be checked by event staff a few times throughout the event. The FIELD is expected to change during MATCH play. Teams can ask the FTA

to re-check specific measurements if they believe something is out of spec prior to a MATCH beginning.

The official drawings, CAD models, and drawings for low-cost versions of important elements of the REBUILT FIELD are posted on [the Playing FIELD webpage](#).

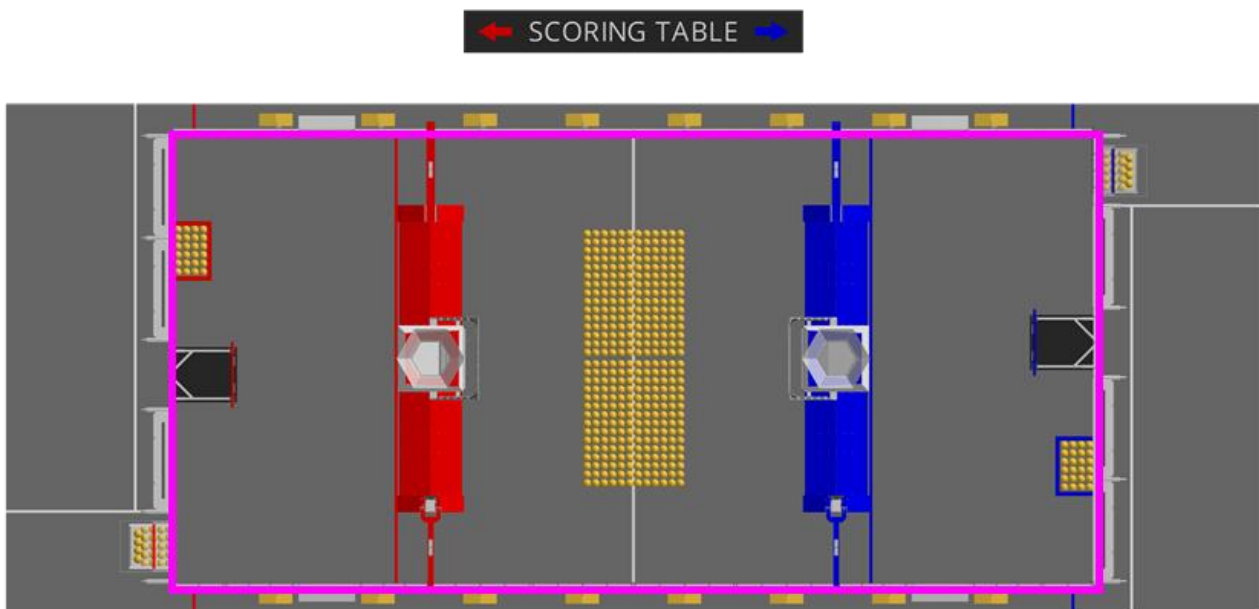
The ARENA is modular and assembled, used, disassembled, and shipped many times during the competition season. It undergoes wear and tear. The ARENA is designed to withstand rigorous play and frequent shipping. Every effort is made to ensure that ARENAS are consistent from event to event.

However, ARENAS are assembled in different venues by different event staff and some small variations occur. Successful teams will design ROBOTS that are insensitive to these variations.

5.2 FIELD

Each FIELD for REBUILT is an approximately 317.7in (~8.07m) by 651.2in (~16.54m) carpeted area bounded by inward facing surfaces of the ALLIANCE WALLS, OUTPOSTS, TOWER WALLS, and guardrails.

Figure 5-2 FIELD boundary in pink



The FIELD is populated with and surrounded by the following elements:

- 1 OUTPOST per ALLIANCE,
- 1 HUB per ALLIANCE,
- 1 TOWER per ALLIANCE,
- 2 DEPOTS,
- 4 BUMPS, and

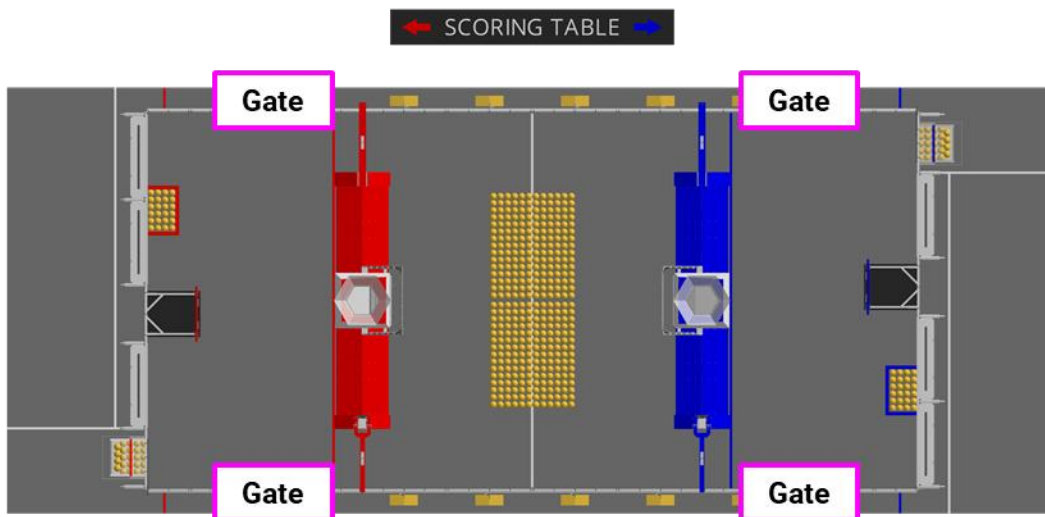
– 4 TRENCHES.

The surface of the FIELD is low pile carpet, Shaw Floors, Philadelphia Commercial, Neyland II 20, “66561 Medallion.” Neyland II carpet is not available for purchase at this time, and the closest equivalent is [Shaw, Philadelphia Brand, Profusion 20, Style 54933](#); see results from evaluation in [this blog post](#).

Carpet edges and seams are secured with [3M™ Premium Matte Cloth \(Gaffers\) Tape GT2, GT3 or comparable Gaffer’s Tape](#). Tears, rips, and damage to the carpet may be repaired with the same styles of tape and ROBOTS must be prepared to operate on surfaces made of carpet, tape, or combinations of both materials as repairs are made through the course of competition.

Guardrails form the long edges of the FIELD. Guardrails are a 20.0in (50.8cm) tall system of transparent polycarbonate supported on the top and bottom by aluminum extrusion. There are 4 gates in the guardrail that allow access to the FIELD for placement and removal of ROBOTS. The gate passthrough, when open, is 38.0in (96.5cm) wide. Gates are closed and shielded during the MATCH.

Figure 5-3 Gate locations



There are 2 versions of guardrails and DRIVER STATIONS used for competitions. 1 design is the Welded FIELD which is reflected in the [2026 Official FIRST FIELD Drawings & Models](#). The other is designed and sold by AndyMark. [Table 5-1](#) and

[Table 5-2](#) illustrate which areas have each kind of FIELD. While the designs are slightly different, the critical dimensions, performance, and expected user experience between them are the same unless otherwise noted. Detailed drawings for the AndyMark design are posted on the [AndyMark website](#). All illustrations in this manual show the traditional Welded FIELD design.

Table 5-1: District Field Types

District	Field Type
FIRST Chesapeake	AndyMark
FIRST California	Welded

District	Field Type
<i>FIRST</i> in Michigan	Welded
<i>FIRST</i> in Texas	AndyMark
<i>FIRST</i> Indiana Robotics	AndyMark
<i>FIRST</i> Israel	Welded
<i>FIRST</i> Mid-Atlantic	Welded
<i>FIRST</i> North Carolina	AndyMark
<i>FIRST</i> South Carolina	Welded
<i>FIRST</i> Wisconsin	AndyMark
NE <i>FIRST</i>	AndyMark
Ontario	Welded
Pacific Northwest	Welded
Peachtree	Welded

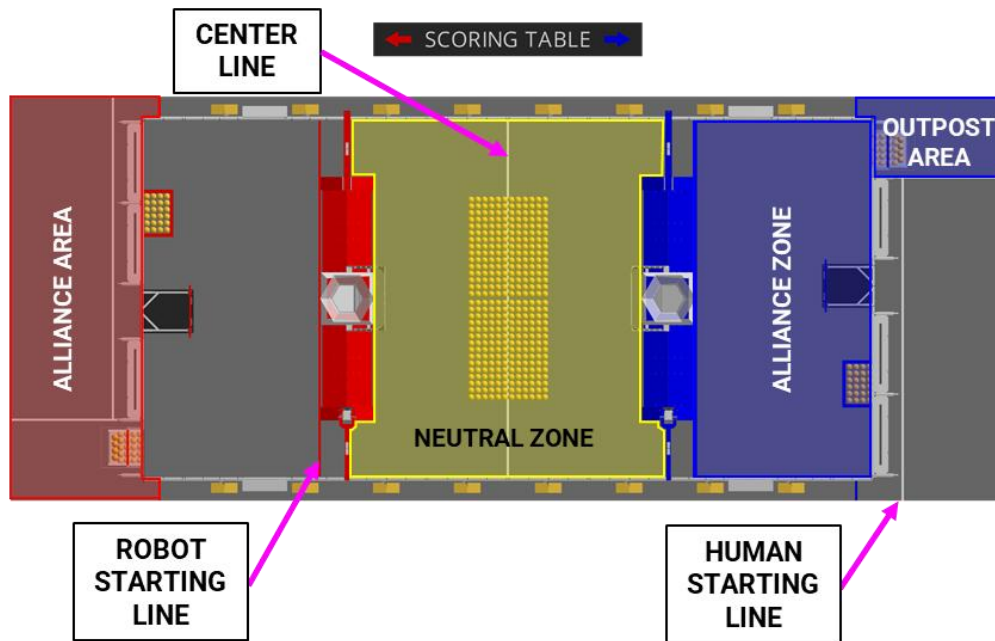
Table 5-2 Regional Field Types

Regional Location	Field Type
Australia	Welded
Brazil	AndyMark
Canada	Welded
China	AndyMark
Mexico	AndyMark
Türkiye	AndyMark
United States	Welded

5.3 Areas, Zones, & Markings

FIELD areas, zones, and markings of consequence are described below. Unless otherwise specified, the tape used to mark lines and zones throughout the FIELD is 2.0in (5.1cm) [3M™ Premium Matte Cloth \(Gaffers\) Tape \(GT2\)](#), [ProGaff® Premium Professional Grade Gaffer Tape](#), or comparable gaffers tape.

Figure 5-4 Areas, markings, and zones



- **ALLIANCE AREA:** an approximately 360in wide by 134in deep (~9.14m by 3.4m) infinitely tall volume formed by, and including the ALLIANCE WALL, OUTPOST, TOWER WALL, the edge of the carpet, and ALLIANCE colored tape perpendicular to the DRIVER STATIONS.
- **ALLIANCE ZONE:** A 158.6in deep by 317.7in long (~4.03m by 8.07m), infinitely tall volume formed by an ALLIANCE WALL, TOWER WALL, and guardrails. It surrounds an ALLIANCE TOWER and a DEPOT. It is bounded by and includes the ROBOT STARTING LINE.
- **CENTER LINE:** a white line that spans the width of the FIELD that bisects the NEUTRAL ZONE in half.
- **NEUTRAL ZONE:** A 283in deep by 317.7in long (7.19m by 8.07m), infinitely tall volume formed by the BUMPS, TRENCHES, HUBS, and guardrails. It surrounds and includes the CENTER LINE.
- **HUMAN STARTING LINE:** a white line spanning the ALLIANCE AREA up to the OUTPOST AREA that is parallel to and located 24.0in (61.0cm) from the bottom square tube of the ALLIANCE WALL to the near edge of the tape.
- **OUTPOST AREA:** a 71.0in wide by 134in deep (1.8m by 3.4m) infinitely tall volume bounded by the OUTPOST, edge of carpet, and ALLIANCE and white colored tape.
- **ROBOT STARTING LINE:** an ALLIANCE colored line that spans the width of the FIELD at the edge of an ALLIANCE ZONE in front of two BUMPS and an ALLIANCE HUB.

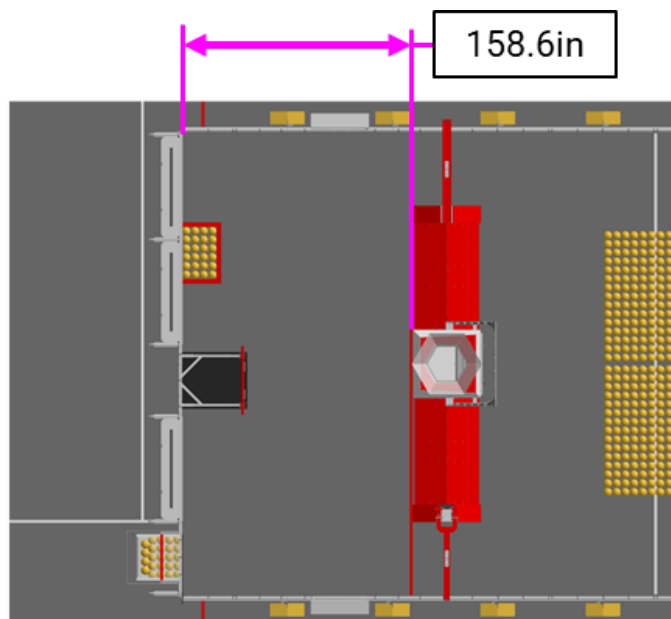
5.4 HUB

Figure 5-5 HUB



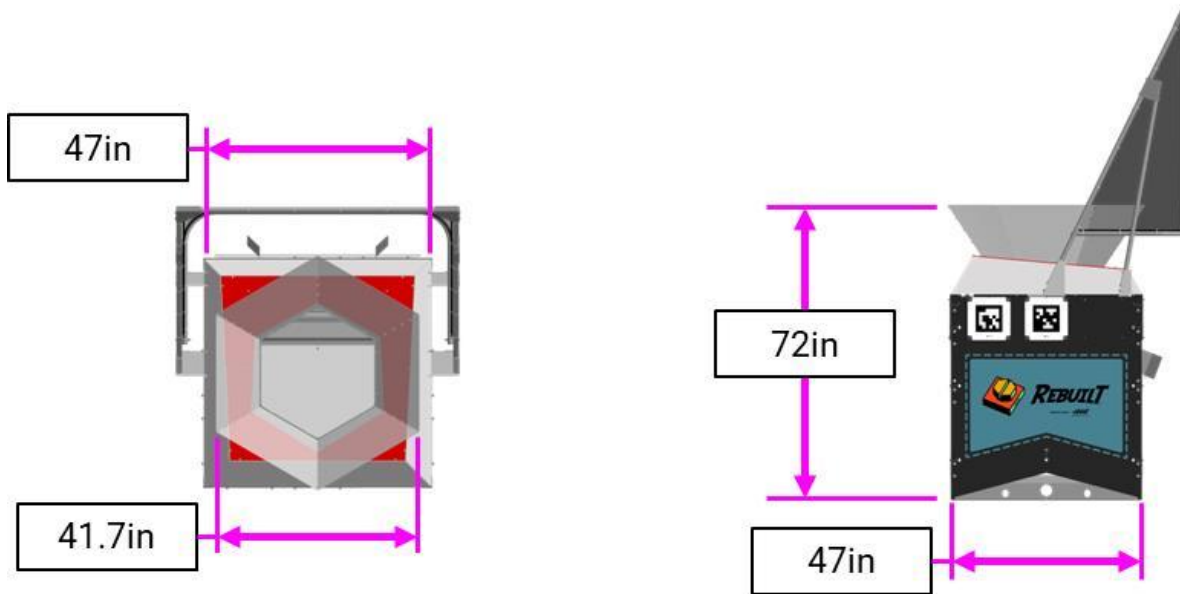
A HUB is one of two 47in by 47in (~1.19m by 1.19m) rectangular prism structures with an extended opening at the top surface. Each ALLIANCE has a dedicated HUB centered between two BUMPS located 158.6in (~4.03m) away from their ALLIANCE WALL. Each HUB has a set of exits that randomly distributes FUEL into the NEUTRAL ZONE. A net structure located in the back of the HUB prevents FUEL launched from most prohibited areas from entering the opening.

Figure 5-6: HUB distance to the ALLIANCE WALL



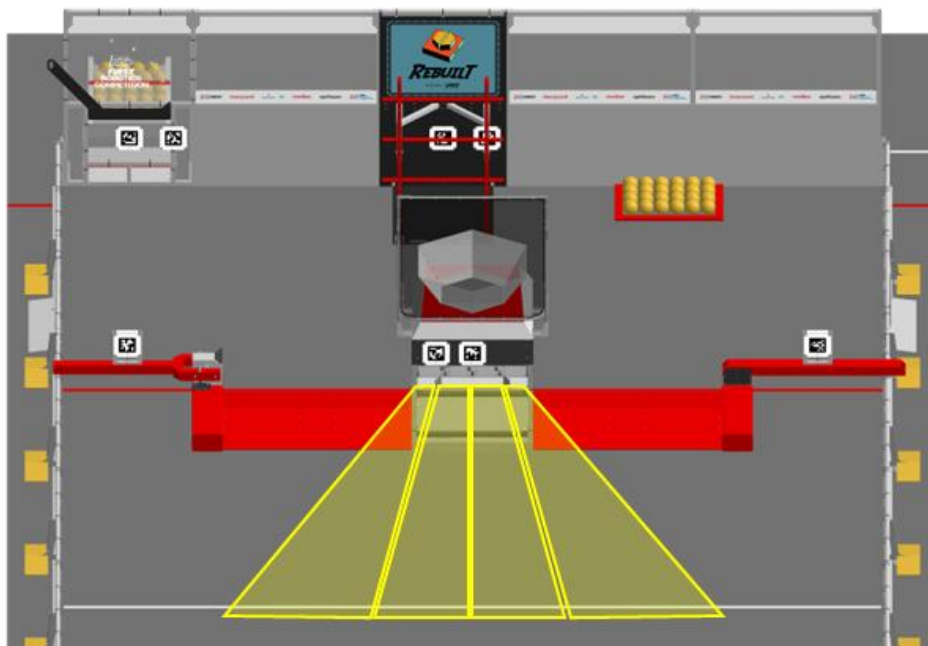
The top of each HUB has a 41.7in (~1.06m) hexagonal opening into which ROBOTS can deliver FUEL. The front edge of the opening is 72in (~1.83m) off the carpet.

Figure 5-7: HUB Dimensions



HUBS have a series of exits at the base of the HUB facing towards the NEUTRAL ZONE. FUEL processed through the HUB are distributed into the NEUTRAL ZONE via one of four exits as shown in [Figure 5-8](#). Examples of FUEL distribution from the HUB can be found on the [Playing FIELD webpage](#).

Figure 5-8: HUB exits (approximation)



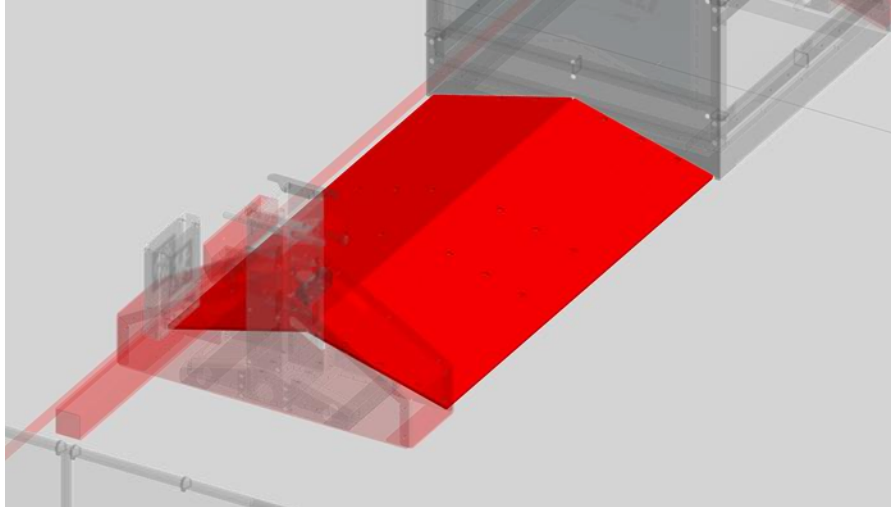
The top angles of the HUB are lit by DMX light bars that indicate if the HUB is active. See [Table 5-3](#) for more details about the various light states in the HUB.

Table 5-3: HUB Lighting

Color	Pre-MATCH	MATCH	Post-Match
ALLIANCE color at 100% brightness	N/A	HUB active	N/A
ALLIANCE color pulsing		HUB deactivation warning. Starts 3 seconds before and continues until deactivation or end of MATCH.	
ALLIANCE color with white chase		During the TRANSITION SHIFT, Indicates the ALLIANCE HUB that will be inactive in ALLIANCE SHIFT 1. HUB is active.	
Purple		N/A	FIELD is safe for FIELD STAFF.
Green			FIELD is safe for all.
White			Post MATCH 3 Second Scoring Assessment Period
Off	MATCH ready to start.	HUB is not active.	N/A

5.5 BUMP

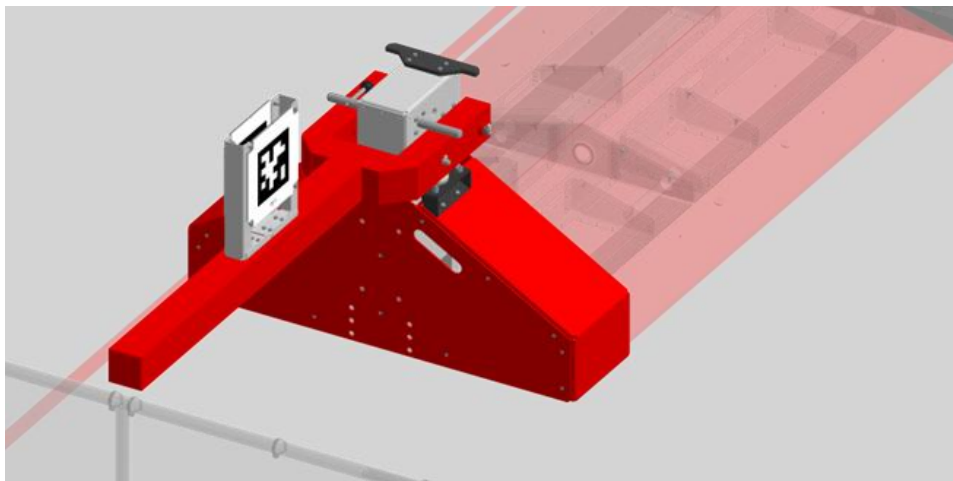
Figure 5-9 BUMP



BUMPS are 73.0in (1.854m) wide, 44.4in (1.128m) deep, and 6.513in (16.54cm) tall structures on either side of the HUB that ROBOTS drive over. The top surface of each BUMP is made up of 0.5in (1.27cm) thick, ALLIANCE colored, Orange Peel textured, HDPE ramps at a 15-degree angle with one ramp sloping down towards the NEUTRAL ZONE and the other ramp sloping down towards the ALLIANCE ZONE.

5.6 TRENCH

Figure 5-10: TRENCH

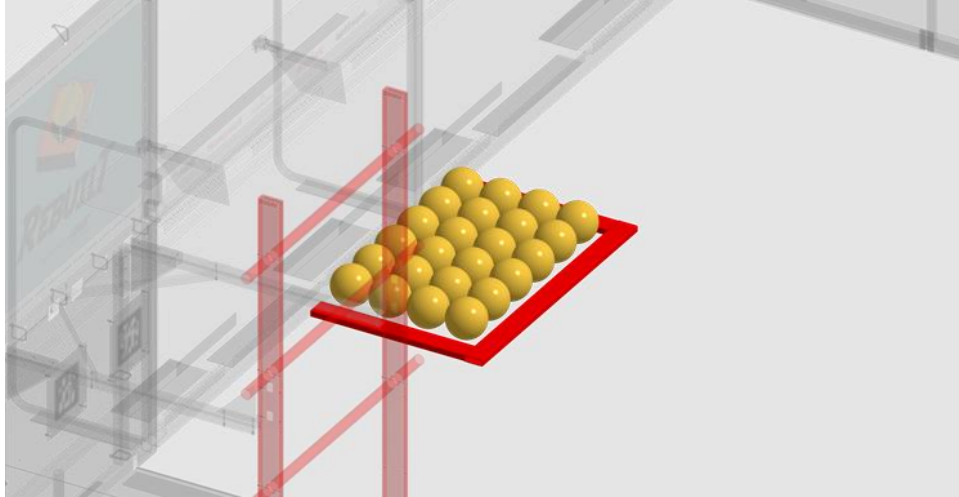


TRENCHES are a 65.65in (1.668m) wide, 47.0in (1.194m) deep, and 40.25in (1.022m) tall structure that ROBOTS drive underneath. The TRENCH extends from the guardrail to the BUMP on both sides of the FIELD. The space underneath each TRENCH arm is 50.34in (1.279m) wide, 22.25in (56.52cm) tall.

TRENCHES along the guardrail closest to the scoring table contain additional electronics to reach the HUB. The TRENCHES along the guardrail furthest from the scoring table have a pivot arm that allows the horizontal portion of the TRENCH to rotate into a vertical position for post-MATCH ROBOT retrieval and to let FIELD staff reset the field between matches. The pivot arm will be locked in the horizontal position during the MATCH.

5.7 DEPOT

Figure 5-11: DEPOT



A DEPOT is a 42.0in (1.07m) wide, 27.0in (68.6cm) deep structure located along the ALLIANCE WALL. There is 1 DEPOT per ALLIANCE. DEPOTS are made up of 3.0in (7.62cm) wide, 1.0in (2.54cm) tall steel barriers. The DEPOT is secured to the carpet using hook fastener which increases the height to approximately 1.125in (2.86cm).

5.8 TOWER

Figure 5-12: TOWER



A TOWER is a 49.25in (1.251m) wide, 45.0in (1.143m) deep, and 78.25in (1.988m) tall structure made up of the TOWER WALL, TOWER BASE, UPRIGHTS, RUNGS and supporting structures. There is 1 TOWER per ALLIANCE. A TOWER is integrated into each ALLIANCE WALL between DRIVER STATION 2 and DRIVER STATION 3.

The TOWER BASE is a 39.0in (99.06cm) wide by 45.18in (1.148m) deep plate that sits on the floor and extends from the TOWER WALL. The TOWER BASE is powder-coated steel with hook fastener underneath. The edges of the TOWER BASE are approximately 0.2in (0.5cm) to 0.3in (0.8cm) tall.

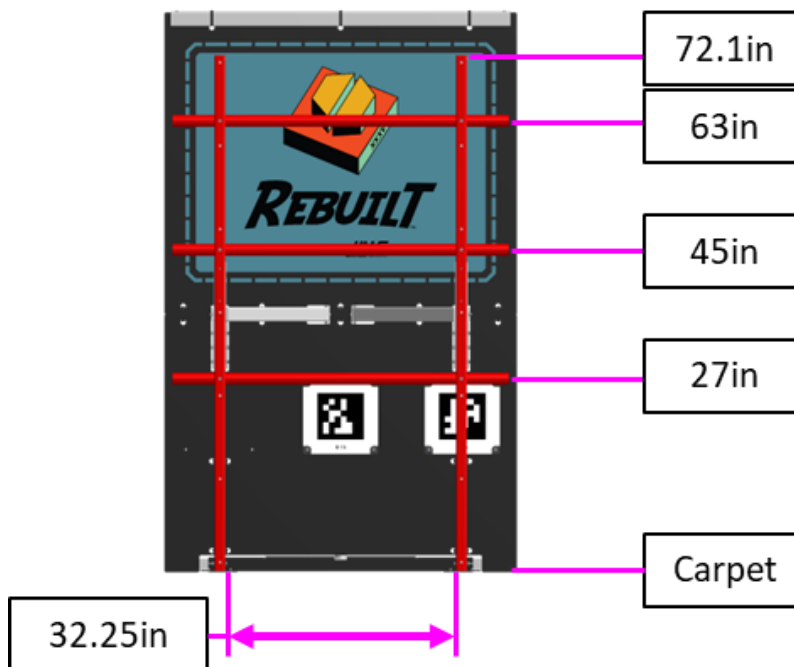
The UPRIGHTS are two 72.1in (1.831m) tall, 1.5in (3.81cm) thick, 3.5in (8.89cm) deep sheet metal box frames that extend vertically up from the TOWER BASE. The distance between each UPRIGHT is 32.25in (81.92cm).

The UPRIGHTS hold three horizontal RUNGS made up of 1-1/4in Sch 40 (1.66in (4.216cm) OD) pipe. Each RUNG is centered between the UPRIGHT and extend 5.875in (14.92cm) from the outer face of the UPRIGHT on either side. The center of the LOW RUNG is located 27.0in (68.58cm) from the floor. The center of the MID RUNG is located 45.0in (114.3cm) from the floor. The center of the HIGH RUNG is 63.0in (1.6m) from the floor. The RUNGS are 18.0in (45.72cm) apart center to center.

The UPRIGHTS and RUNGS are powder-coated red or blue.

Each TOWER has additional supporting structures extending from the UPRIGHT to the TOWER WALL between approximately 28.40in (72.14cm) and 43.38in (1.102m) off the floor.

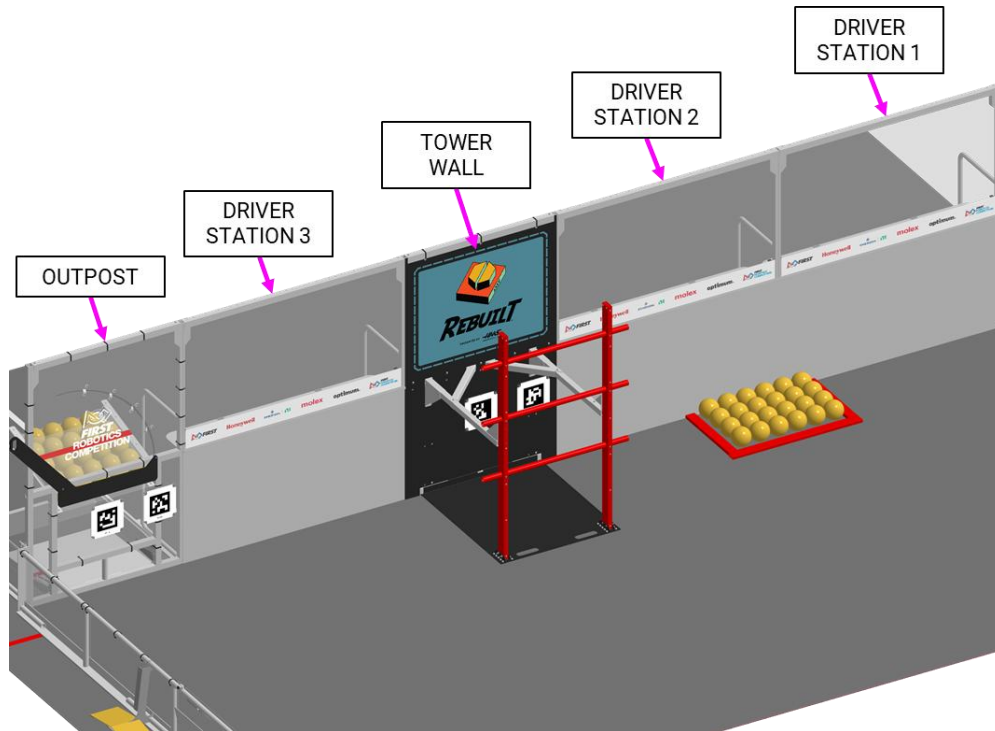
Figure 5-13: TOWER Dimensions



5.9 ALLIANCE WALL

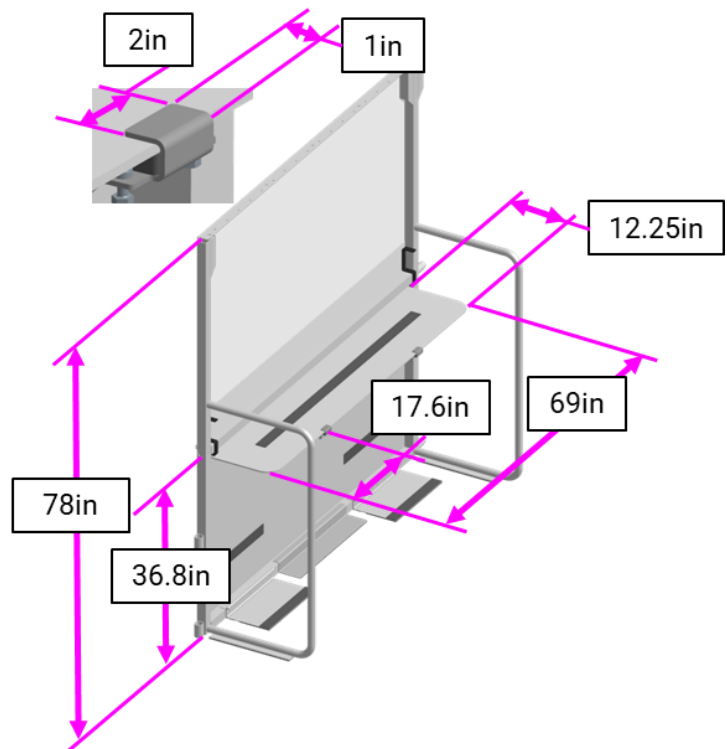
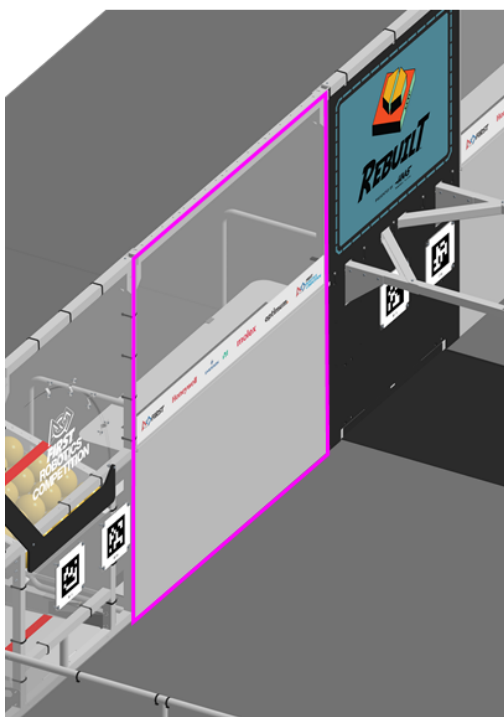
The ALLIANCE WALL separates ROBOTS from DRIVE TEAM members in the ALLIANCE AREA. It consists of 3 DRIVER STATIONS, an OUTPOST, and a TOWER WALL.

Figure 5-14: ALLIANCE WALL



5.9.1 DRIVER STATIONS

Figure 5-15: DRIVER STATION (ALLIANCE AREA perspective and FIELD perspective)



A DRIVER STATION is 1 of 3 assemblies within an ALLIANCE WALL behind which a DRIVE TEAM operates their ROBOT. Each DRIVER STATION is made from a 36.8in (93.5cm) tall diamond plate base topped with a 42in (1.07m) tall transparent plastic sheet and a top rail.

An aluminum shelf is attached to each DRIVER STATION to support an OPERATOR CONSOLE. The shelf is 69in (1.75m) wide and 12.25in (31.1cm) deep. There is a 54.0in (1.372m) long by 2.0in (nominal) wide strip of hook-and-loop tape ("loop" side) along the center of the support shelf that may be used to secure the OPERATOR CONSOLE to the shelf. The shelf also includes two clips to hold the shelf in place with a 1.0in (2.5cm) wide by 2.0in (5.1cm) deep tab that sits on the top surface of the shelf.

There is a 6.0in (15.2cm) tall sponsor panel in each DRIVER STATION. The top of this panel is 42.9in (1.09m) from the carpet.

There may be a ramp available at events for DRIVE TEAMS with limited mobility. It is designed to allow an individual using a wheelchair to access the DRIVER STATION shelf and/or see onto the FIELD; however, this accommodation is available to anyone who has a wheelchair or other physical disability that obstructs their view of the FIELD. Teams should speak to the FTA before MATCHES begin to ensure that it is available for each of the team's MATCHES.

This ramp is available at many Regional and District events. For questions, please connect with the local [Program Delivery Partner](#).

Teams should also speak to the FTA for any other FIELD side accommodations needed.

Each DRIVER STATION contains the following elements for DRIVE TEAMS:

- 1 Ethernet cable: attaches to the Ethernet port of the OPERATOR CONSOLE and provides connectivity to the FIELD Management System (FMS).
- 1 120VAC NEMA 5-15R power outlet (i.e. standard US outlet): located on each DRIVER STATION shelf and protected by its own 2-Amp circuit breaker. It can be used to power the OPERATOR CONSOLE. DRIVE TEAMS are responsible for monitoring their power consumption as a tripped breaker in the outlet does not constitute an ARENA FAULT. For some events in regions that don't use NEMA 5-15 shaped outlets, event organizers may install appropriate plug adapters to be used throughout the event.
- 1 Emergency Stop (E-Stop) button: located on the left side of the DRIVER STATION shelf and is used to deactivate a ROBOT in an emergency.
- 1 Autonomous Stop (A-Stop) button: located on the right side of the DRIVER STATION shelf and is used to DISABLE a ROBOT during AUTO.
- 1 team sign: located at the top of each DRIVER STATION. The FIELD facing side of the sign displays the team number in the ALLIANCE color. The ALLIANCE AREA side of the sign displays the following information in red:
 - Pre-MATCH: team number and ROBOT connection state
 - During Qualification MATCHES:
 - Current SHIFT and time remaining in that period,
 - A shows for AUTO when both ALLIANCE's HUBS are active
 - T shows for the TRANSITION SHIFT when both ALLIANCE's HUBS are active
 - R shows when the Red ALLIANCE'S HUB is active

- B shows when the Blue ALLIANCE'S HUB is active
- E shows for END GAME when both ALLIANCE's HUBS are active
- Progress towards the FUEL Ranking Points. This shows total scored out of the ENERGIZED RP and once that threshold passes it shows out of the SUPERCHARGED RP
- AUTO TOWER points, and
- remaining MATCH period time.

Figure 5-16 Back of team sign (during Qualification MATCH)



- During the MATCH during Playoff MATCHES:
 - Which MATCH period is active and time remaining in that period,
 - MATCH scores, and
 - remaining MATCH period time.

Figure 5-17 Back of team sign (during Playoff MATCH)



- 1 timer (in DRIVER STATION 2 only): displays the official time remaining in the MATCH period on the FIELD-facing side (in white) and on the team facing side the following information in red:
 - remaining MATCH period time, and
 - MATCH scores.

Figure 5-18 Back of timer



- 1 team LED stack: indicates ALLIANCE color, ROBOT status, E-Stop and A-Stop status, and is centered at the top of each DRIVER STATION.

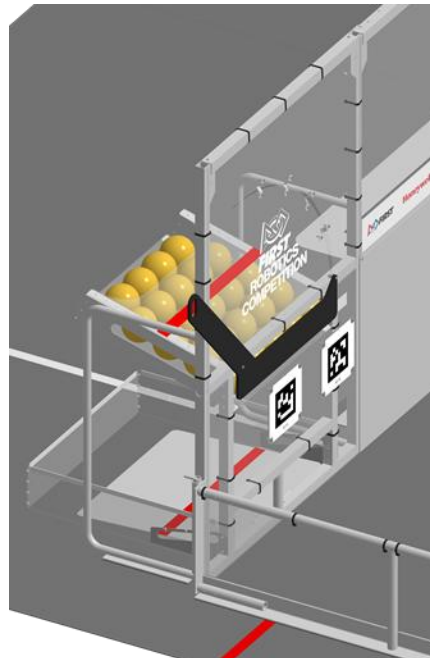
The stack includes 2 identical ALLIANCE-colored ROBOT status LEDs above a third amber A-stop/E-stop LED. LED states are as follows:

- ROBOT status LEDs
 - Solid: indicates that the ROBOT is connected and enabled. This only happens during a MATCH.
 - Blinking: indicates that either the FMS is preset for the MATCH and the ROBOT is not connected yet, or it's during a MATCH and the corresponding ROBOT has lost connectivity, or the E-stop was pressed.

- Off: indicates that the ROBOT is linked and DISABLED prior to the start of the MATCH, or the ROBOT is BYPASSED. This light is also off, regardless of ROBOT connection status, after the MATCH has concluded.
- A-Stop/E-stop LED
 - Solid: the ROBOT is DISABLED due to a press of the team E-stop button, the FIELD E-stop button, or by the scorekeeper via the FMS.
 - Blinking: the ROBOT is DISABLED for the remainder of AUTO due to a press of the team A-Stop button.
 - Off: the ROBOT is not DISABLED by the FIELD.
- FMS hardware and wiring: mostly located below shelves in the 2 DRIVER STATIONS closer to the scoring table.

5.9.2 OUTPOST

Figure 5-19 OUTPOST



An OUTPOST is an assembly through which HUMAN PLAYERS feed FUEL into the FIELD and ROBOTS can deliver FUEL to their HUMAN PLAYERS. There are 2 OUTPOSTS, 1 on either end of the FIELD connecting the guardrail to the ALLIANCE WALL. Each OUTPOST has a 31.8in (80.8cm) wide by 7.0in (17.8cm) tall opening through which FUEL passes to the FIELD. The bottom of the opening is 28.1in (71.4cm) off the floor.

A 15.0° sloped tunnel, called the CHUTE, leads to the upper opening in the OUTPOST. The CHUTE can hold approximately 25 FUEL at a time retained by the CHUTE DOOR. The CHUTE DOOR is an HDPE arm on a pivot that can be rotated approximately 90 degrees by the HUMAN PLAYER to open or close the CHUTE.

At the base of the OUTPOST is an opening 32.0in (81.3cm) wide by 7.0in (17.8cm) tall where ROBOTS can push FUEL into the CORRAL. The bottom of the opening is 1.88in (4.77cm) off the ground. The opening is divided in the center by a vertical 1-1/4in Sch 40 (1.66in (4.216cm) OD) pipe. The CORRAL is made up of 8.13in (20.6cm) tall polycarbonate panels that create a 35.8in (90.8cm) wide, 37.6in (95.5cm) deep area on the floor in which FUEL can be stored.

The CHUTE and CORRAL each feature an ALLIANCE-colored tape line that indicates where DRIVE TEAMS may be restricted from reaching. The near edge of the CORRAL tape is 12.7in (32.3cm) from the FIELD-facing wall of the OUTPOST. The near edge of the CHUTE tape line is 12.9in (32.8cm) from the FIELD-facing wall of the OUTPOST.

There are 2 stools available at events for DRIVE TEAMS to use. Each stool is 23.0in (58.42cm) wide by 13.5in (34.29cm) deep, 6.25in (15.88cm) tall, and rated for 300lb (136.0kg).

It is specially intended to allow individuals who are shorter, better sightlines onto the FIELD; however, this accommodation is available to anyone who has another physical disability that obstructs their view of the FIELD.

Only 2 stools are available, and priority will be given to those with the biggest need. Teams should speak to the FTA before MATCHES begin to request that it is available for each of the team's MATCHES. Teams may also purchase their own ([Item Number: 779ac01stpm](#)) or bring an equivalent (e.g. not foldable and similar dimensions) version to the event to guarantee use for every match.

This stool is available at all events within the US & Canada and equivalents are available at international events. For questions, please connect with the local [Program Delivery Partner](#).

5.10 SCORING ELEMENTS

SCORING ELEMENTS are items that teams use to score points. There is one type of SCORING ELEMENT used in REBUILT: FUEL.

In REBUILT, a ROBOT may CONTROL any number of SCORING ELEMENTS after the start of the MATCH.

5.10.1 FUEL

Figure 5-20 FUEL



A FUEL is a 5.91in (15.0cm) diameter, high density foam ball. FUEL is a custom made SCORING ELEMENT available for purchase on the AndyMark website, [am-5801](#).

FUEL has a weight of between 0.448-0.500lb (~0.203-0.227kg).

FUEL undergoes wear and tear during a competition. Generally, a FUEL that still appears to look approximately like a FUEL is considered a FUEL for the purposes of rule evaluation and scoring, whether damaged or not.

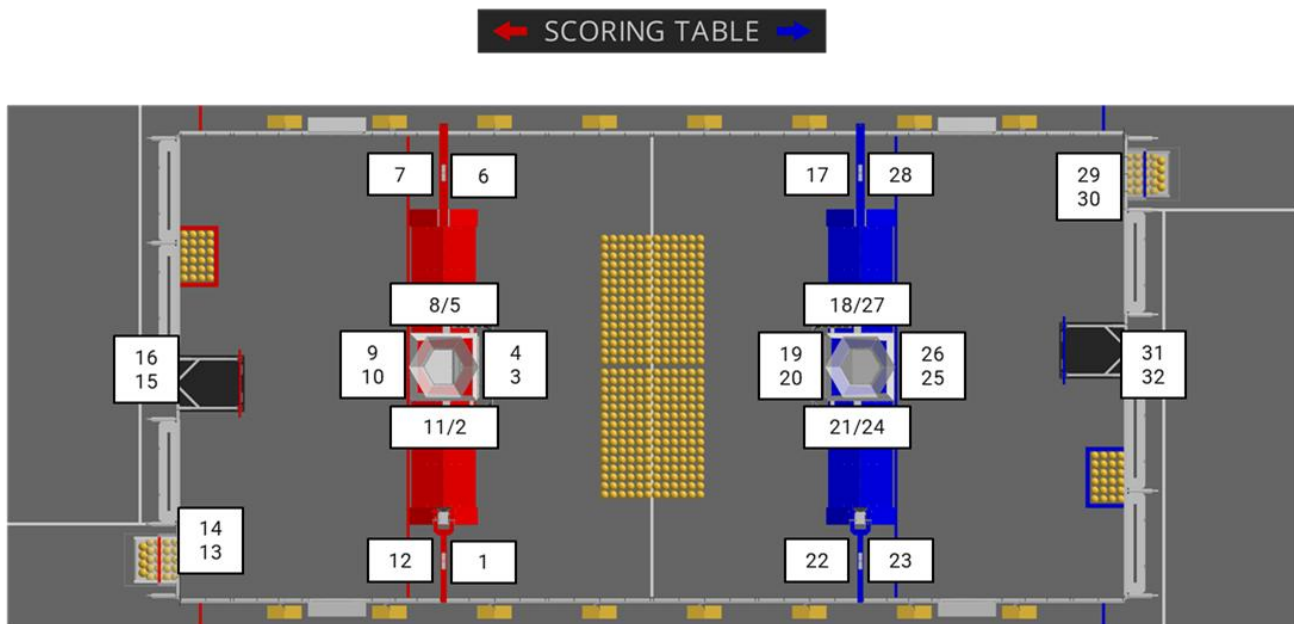
Small chunks of a FUEL are not considered a FUEL.

Most new FUEL will have small impressions due to being pressed against other FUEL or the sides of the packaging during shipping. These impressions are purely cosmetic and do not affect the function of the FUEL. Most impressions will gradually disappear as the FUEL is used.

5.11 AprilTags

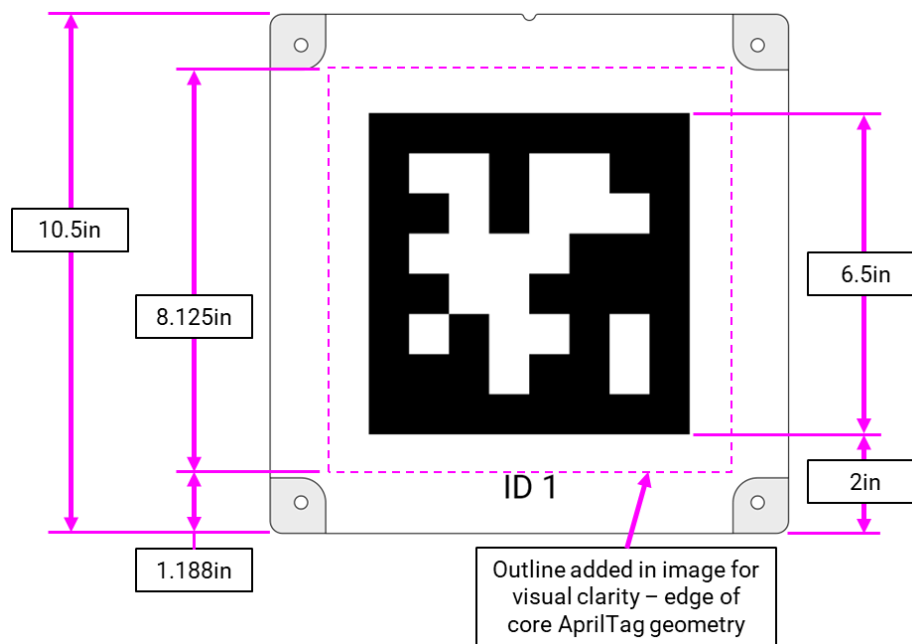
AprilTags are 8.125in (20.64cm) square targets located on the HUB, TOWER WALL, OUTPOST, and TRENCHES. There are 32 unique markers on the FIELD positioned as shown in [Figure 5-21](#).

Figure 5-21 AprilTag Locations



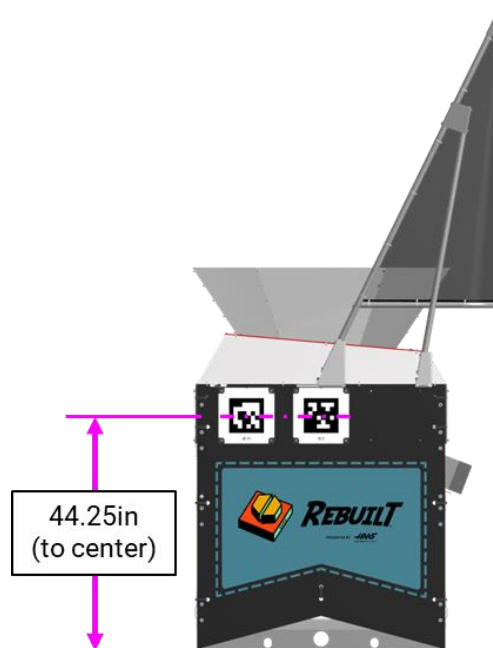
All markers are from the 36h11 tag family, IDs 1-32. All AprilTags are mounted to and centered on a 10.5in (26.67cm) square polycarbonate panel. Each marker has an identifying text label. If AprilTags experience wear and marking during MATCHES they are repaired with gaffers tape.

Figure 5-22 AprilTag sizing



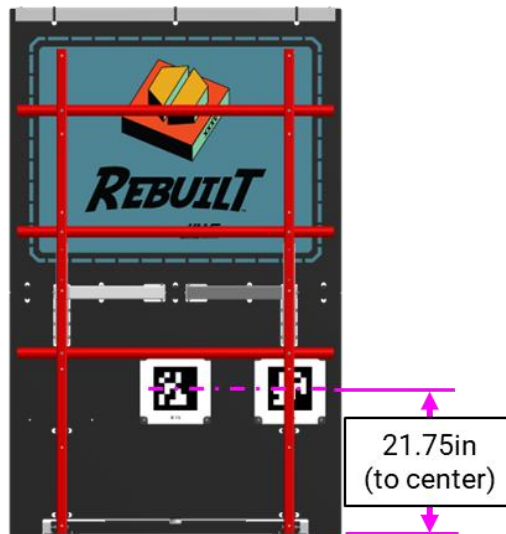
HUB AprilTags (IDs 2, 3, 4, 5, 8, 9, 10, 11, 18, 19, 20, 21, 24, 25, 26, 27) are located on all four faces of the HUB. Each face has two AprilTags with centers located 44.25in (1.124m) off the floor as shown in [Figure 5-23](#). One AprilTag per face is centered and the other AprilTag is horizontally offset.

Figure 5-23 HUB AprilTags



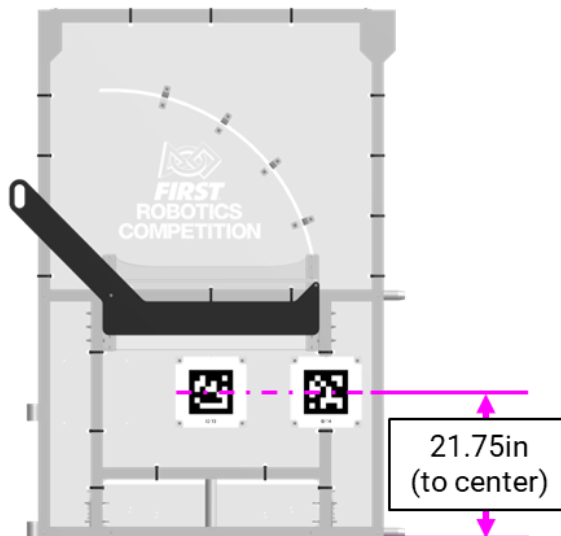
Two AprilTags (IDs 15, 16, 31, 32) are located on each TOWER WALL with centers 21.75in (55.25cm) off the floor as shown in [Figure 5-24](#). One AprilTag per TOWER is centered and the other AprilTag is horizontally offset.

Figure 5-24 TOWER WALL AprilTags



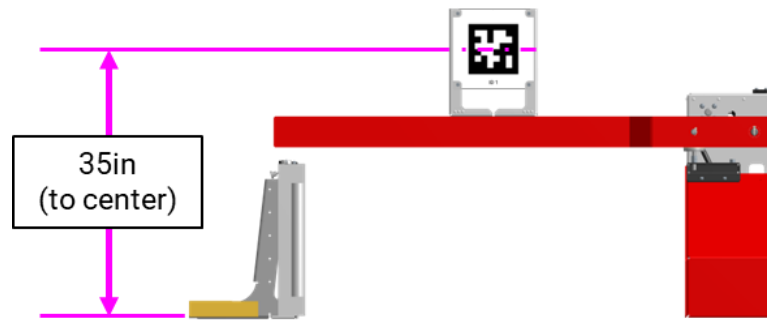
Two AprilTags (IDs 13, 14, 29, 30) are located on each OUTPOST with centers 21.75in (55.25cm) off the floor as shown in [Figure 5-25](#). One AprilTag per OUTPOST is centered with the CHUTE and CORRAL openings and the other AprilTag is horizontally offset.

Figure 5-25 OUTPOST AprilTags



TRENCH AprilTags (IDs 1, 6, 7, 12, 17, 22, 23, 28) are attached to mounting brackets located on the top surface of the horizontal arm of the TRENCH. Each TRENCH has two AprilTags, one facing the ALLIANCE ZONE and one facing the NEUTRAL ZONE. TRENCH AprilTags are approximately centered on the opening under the TRENCH arm and their centers are located 35in (88.9cm) off the floor as shown in [Figure 5-26](#).

Figure 5-26 TRENCH AprilTags



For further marker locating information please refer to the [2026 Field Dimension Drawings](#). Printable copies of the field AprilTags can be found on the [Playing FIELD webpage](#).

5.12 The FIELD Management System

The FIELD Management System (FMS) is all the electronics responsible for sensing and controlling the FIRST Robotics Competition FIELD. The FMS encompasses all FIELD electronics, including computers, REFEREE touchscreens, the wireless access point, sensors, stack lights, A-Stops and E-Stops, etc.

When a DRIVE TEAM connects the Ethernet cable from their assigned DRIVER STATION to their OPERATOR CONSOLE, the Driver Station Software on the OPERATOR CONSOLE computer communicates with FMS. Once connected, the open ports available are described in [Table 8-5](#).

Note that ROBOT code cannot be deployed while connected to the FMS. Additional information about the FMS may be found in the [FMS Whitepaper](#).

The FMS alerts participants to milestones in the MATCH using audio cues detailed in [Table 5-4](#). Please note that audio cues are intended as a courtesy to participants and not intended as official MATCH markers. If there is a discrepancy between an audio cue and the FIELD timers, the FIELD timers are the authority.

Table 5-4: Audio Cues

Event	Timer Value(s)	Audio Cue
MATCH start	0:20 (for AUTO)	"Cavalry Charge"
AUTO ends	0:00 (for AUTO)	"Buzzer"
TELEOP & TRANSITION begins	2:20	"3 Bells"
ALLIANCE SHIFT starts	2:10 1:45 1:20 0:55	"POWER UP – Linear Popping"
END GAME begins	0:30	"Steam Whistle"
MATCH end	0:00	"Buzzer"
MATCH stopped	n/a	"Foghorn"

5.13 FIELD STAFF

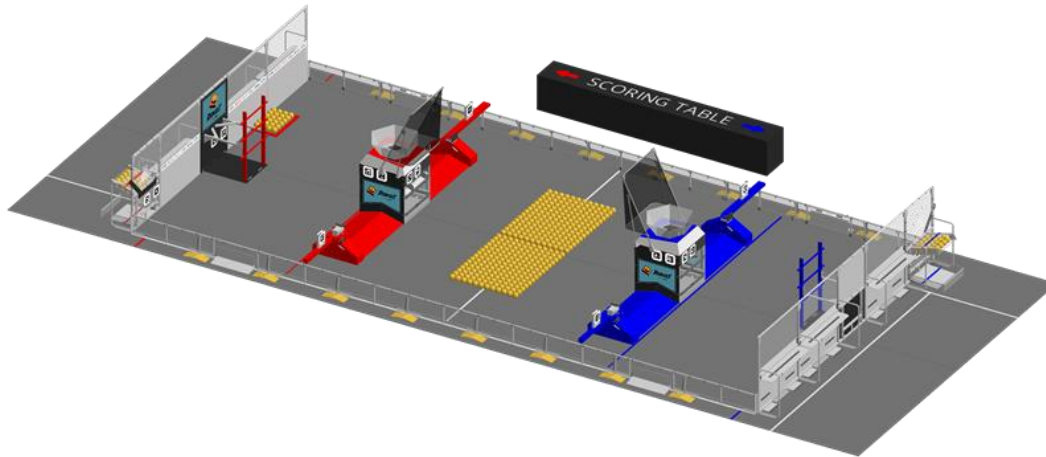
FIELD STAFF are responsible for making sure the MATCHES are cycled through efficiently, fairly, safely, and with a spirit of cooperation, *Gracious Professionalism*, and generosity of spirit. FIELD STAFF roles are filled by volunteers from the community who prepare for the event with thorough training and certification. There are 3 FIELD-side key volunteer roles with whom teams should be familiar and are encouraged to use as resources to make their event experience valuable (in whatever way the team defines “valuable”).

- **Head REFEREE** – trains, directs, and supervises REFEREES. They oversee all scoring processes and procedures in collaboration with the *FIRST* Technical Advisor (FTA). They interact with STUDENTS, volunteers, and contracted/*FIRST* staff. The Head REFEREE is positioned between the FIELD and the scoring table and wears a yellow shirt. The Head REFEREE has final authority for decisions regarding MATCH scores, penalties, and YELLOW and RED CARD assignments. For additional details, please refer to the [Head REFEREE role description](#).
- ***FIRST* Technical Advisor (FTA)** - ensures events run smoothly, safely, and in accordance with *FIRST* requirements. The FTA collaborates with *FIRST* staff, event staff, and other event volunteers in many different areas at events. The FTA is the liaison between *FIRST* HQ and the event for all things related to the FIELD, ROBOTS, and game, acts as a team advocate for all teams competing at the event and is a major point of escalation and conflict resolution for the event. For additional details, please refer to the [FTA role description](#).
- **FIELD Supervisor** - directs activity on the FIELD to ensure efficient execution of the MATCHES, pacing of the event, and smooth flow of MATCH play. FIELD Supervisors are responsible for ensuring the FIELD is intact and lead FIELD Reset teams, who are responsible for resetting the FIELD after each MATCH in preparation for the subsequent MATCH. For additional details, please refer to the [FIELD Supervisor role description](#).



6 Game Details

Figure 6-1 REBUILT



In REBUILT, 2 ALLIANCES (an ALLIANCE is a cooperative of up to 4 *FIRST* Robotics Competition teams) play MATCHES, set up and implemented per the details described below.

6.1 MATCH Overview

MATCHES run on 7–10-minute cycles, which consist of pre-MATCH setup, the 2-minute and 40-second MATCH, and the post-MATCH reset.

During the MATCH, ROBOTS collect FUEL and score in their HUB. HUMAN PLAYERS can deliver FUEL to ROBOTS or score them in their HUB.

ROBOTS conclude the MATCH climbing on their TOWER.

6.2 DRIVE TEAM

A DRIVE TEAM is a set of up to 5 people from the same *FIRST* Robotics Competition team responsible for team performance for a specific MATCH. There are 4 specific roles on a DRIVE TEAM which ALLIANCES can use to assist ROBOTS, and no more than 1 member of the DRIVE TEAM may be a non-STUDENT.

The intent of the definition of DRIVE TEAM and DRIVE TEAM related rules is that, barring extenuating circumstances, the DRIVE TEAM consists of people who arrived at the event affiliated with that team and are responsible for their team's and ROBOT'S performance at the event (this means a person may be affiliated with more than 1 team).

The intent is not to allow teams to “adopt” members of other teams for strategic advantage for the loaning team, borrowing team, and/or their ALLIANCE (e.g. an ALLIANCE CAPTAIN believes 1 of their DRIVERS has more experience than a DRIVER on their first pick, and the teams agree the first pick team will “adopt” that DRIVER and make them a member of their DRIVE TEAM for Playoffs).

The definition isn't stricter for 2 main reasons. First, to avoid additional bureaucratic burden on teams and event volunteers (e.g. requiring that teams submit official rosters that Queuing must check before allowing a DRIVE TEAM into the ARENA). Second, to provide space for exceptional circumstances that give teams the opportunity to display *Gracious Professionalism* (e.g. a bus is delayed, a DRIVE COACH has no DRIVERS, and their pit neighbors agree to help by loaning DRIVERS as temporary members of the team until their bus arrives).

Table 6-1 DRIVE TEAM roles

Role	Description	Max./	Criteria
DRIVE TEAM			
DRIVE COACH	a guide or advisor	1	any team member, must wear "DRIVE COACH" button
TECHNICIAN	a resource for ROBOT troubleshooting, setup, and removal from the FIELD	1	any team member, must wear "TECHNICIAN" button
DRIVER	an operator and controller of the ROBOT	3	STUDENT, must wear a "DRIVE TEAM" button
HUMAN PLAYER	a SCORING ELEMENT manager		

A STUDENT is a person who has not completed high-school, secondary school, or the comparable level as of September 1 prior to Kickoff.

The TECHNICIAN provides teams with a technical resource for pre-MATCH setup, ROBOT connectivity, OPERATOR CONSOLE troubleshooting, and post-MATCH removal of the ROBOT. Some pre-MATCH responsibilities for the TECHNICIAN may include, but are not limited to:

- location of the ROBOT radio, its power connection, and understanding of its indicator lights
- location of the roboRIO and understanding of its indicator lights
- username and password for the OPERATOR CONSOLE
- restarting the Driver Station and Dashboard software on the OPERATOR CONSOLE
- changing the bandwidth utilization (e.g. camera resolution, frame rate, etc.)
- changing a battery
- charging pneumatics

While the TECHNICIAN may be the primary technical member of the DRIVE TEAM, all members of the DRIVE TEAM are encouraged to have knowledge of the basic functionality of the ROBOT, such as the location and operation of the main circuit breaker, connecting and resetting joysticks or gamepads from the OPERATOR CONSOLE, and removing the ROBOT from the FIELD.

Per [Table 6-1](#), each team is allotted up to 3 STUDENTS who may be assigned as DRIVERS or HUMAN PLAYERS. If multiple teams on an ALLIANCE do not have 3 STUDENTS at the event to act in these roles, 1 of

the ALLIANCE'S teams may substitute a STUDENT TECHNICIAN as a HUMAN PLAYER for that MATCH only. In this case,

- the Head REFEREE must be notified,
- all HUMAN PLAYER rules now apply to this DRIVE TEAM member, and
- this DRIVE TEAM member is no longer considered a TECHNICIAN for that MATCH.

6.3 Setup

Before each MATCH begins, FIELD STAFF stage SCORING ELEMENTS as described in section [6.3.4 SCORING ELEMENTS](#). DRIVE TEAMS stage their ROBOTS (as described in section [6.3.3 ROBOTS](#)) and OPERATOR CONSOLES (as described in section [6.3.2 OPERATOR CONSOLES](#)). Then, DRIVE TEAMS take their places as described in section [6.3.1 DRIVE TEAMS](#).

6.3.1 DRIVE TEAMS

DRIVE TEAMS prepare for a MATCH by staging in the appropriate areas, according to their role on the DRIVE TEAM, and by identifying themselves accordingly. DRIVE TEAM starting conditions are listed below, and a DRIVE TEAM obstructing or delaying any of the conditions is at risk of violating [G301](#).

- only DRIVE TEAM members assigned to the upcoming MATCH are present,
- for Qualification MATCHES only DRIVE TEAM members whose ROBOTS have passed initial, complete Inspection are present,
- any number of HUMAN PLAYERS are staged in their OUTPOST AREAS,
- HUMAN PLAYERS not included in C, DRIVERS, and DRIVE COACHES are in their ALLIANCE AREA and behind the HUMAN STARTING LINE,
- TECHNICIANS are in the event-designated area near the FIELD,
- DRIVE TEAM members clearly display their designated buttons above their waists (DRIVE COACH - "Drive Coach", DRIVERS and HUMAN PLAYERS - "Drive Team", and TECHNICIAN - "Technician"), and
- if a Playoff MATCH, the ALLIANCE CAPTAIN clearly displays their designated ALLIANCE CAPTAIN identifier (e.g. hat or armband).

6.3.2 OPERATOR CONSOLES

DRIVE TEAMS set up their OPERATOR CONSOLE as soon as the DRIVE TEAM from the previous MATCH has cleared the area. OPERATOR CONSOLES must be compliant with all relevant rules, specifically those in section [8.9 OPERATOR CONSOLE](#). The OPERATOR CONSOLE is plugged into the team's assigned DRIVER STATION, as indicated on the team sign. Any control devices worn or held by their HUMAN PLAYERS and/or DRIVERS during the MATCH must be disconnected from or set on or beside the OPERATOR CONSOLE before the MATCH can begin. A DRIVE TEAM obstructing or delaying OPERATOR CONSOLE setup is at risk of violating [G301](#).

For the purposes of FIRST Robotics Competition, any device connected to the OPERATOR CONSOLE is considered a control device because REFEREES are not expected to differentiate between devices that can or cannot control the ROBOT.

6.3.3 ROBOTS

DRIVE TEAMS stage their ROBOT in accordance with [G303](#). A DRIVE TEAM obstructing or delaying ROBOT setup requirements is at risk of violating [G301](#).

If order of placement matters to either or both ALLIANCES, the ALLIANCE notifies the Head REFEREE during setup for that MATCH, and the Head REFEREE instructs ALLIANCES to alternate placement of ROBOTS. In a Qualification MATCH, REFEREE instructions are that ROBOTS are placed in the following order:

1. red DRIVER STATION 1 ROBOT
2. blue DRIVER STATION 1 ROBOT
3. red DRIVER STATION 2 ROBOT
4. blue DRIVER STATION 2 ROBOT
5. red DRIVER STATION 3 ROBOT
6. blue DRIVER STATION 3 ROBOT

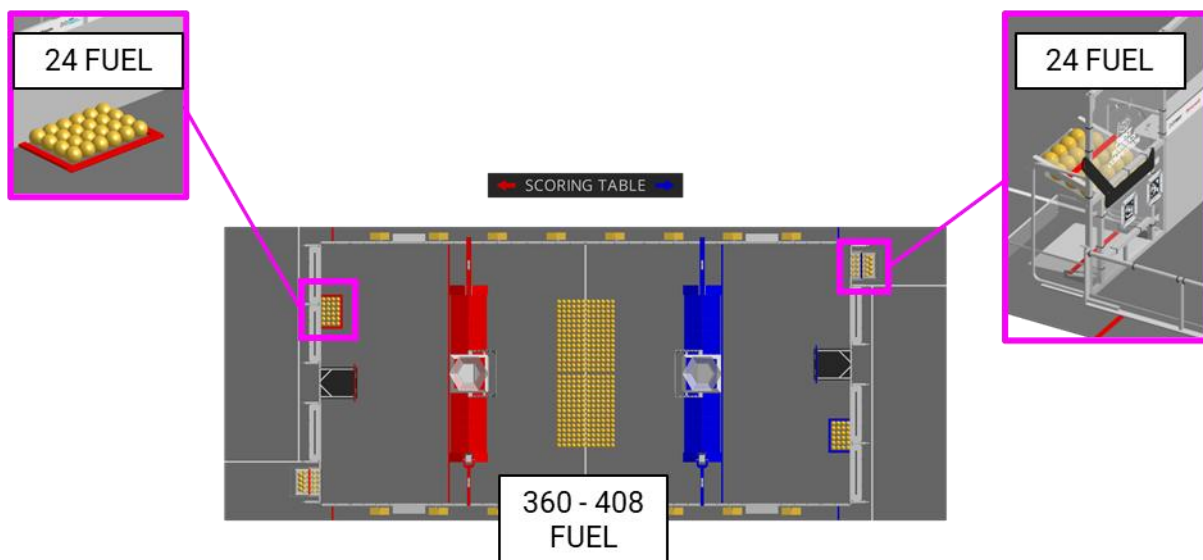
In an intra-Division Playoff MATCH, the same pattern is applied, but instead of blue ALLIANCE placing last, the higher seeded ALLIANCE (regardless of color) places last. For inter-Division Playoff MATCHES, the ALLIANCE that places last is determined by a (real or virtual) coin flip facilitated by the Head REFEREE where a “heads” result invites the red ALLIANCE to place last.

FIELD STAFF may ask teams to indicate their intended location and are not required to wait for a team to stage their ROBOT in its exact location before moving to the next team.

6.3.4 SCORING ELEMENTS

Before each MATCH begins, FIELD STAFF stage SCORING ELEMENTS.

Figure 6-2 SCORING ELEMENT staging positions



504 FUEL are staged for each MATCH as follows (reference [Figure 6-2](#)):

- A. 24 FUEL are staged in each DEPOT (FUEL may not be in a uniform layout),
- B. 24 FUEL are staged in each OUTPOST CHUTE,
- C. 8 FUEL may be preloaded in each ROBOT by the ROBOT'S DRIVE TEAM, such that each FUEL is fully supported by the ROBOT (up to 48 total; a FUEL not pre-loaded in a ROBOT is staged in the NEUTRAL ZONE), and
- D. remaining FUEL are arranged in the NEUTRAL ZONE through a process listed in Section 6.3.4.1 (360 to 408, depending on how many are preloaded in ROBOTS).

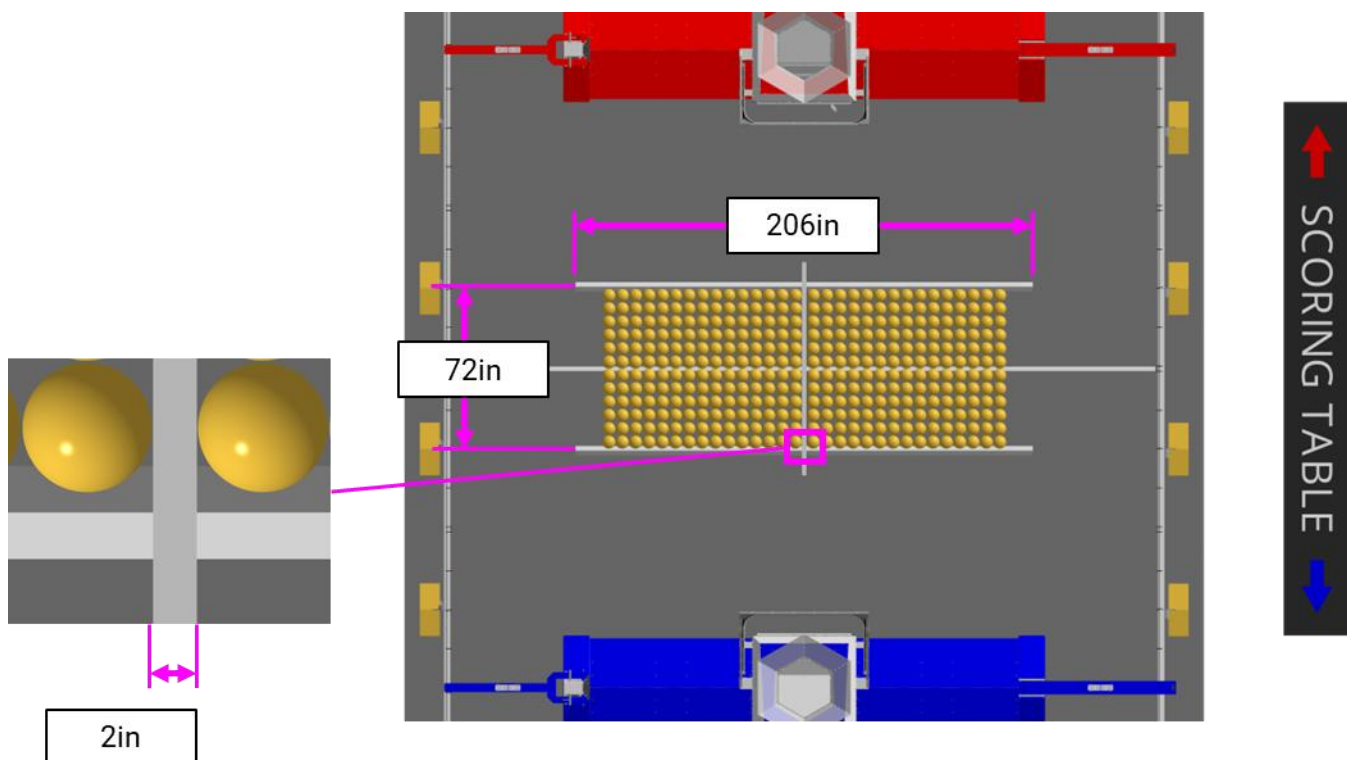
Due to the high number of SCORING ELEMENTS in REBUILT, FIELD STAFF may not be able to perfectly count all FUEL put into the NEUTRAL ZONE. The number of FUEL staged in the NEUTRAL ZONE for any given match may fluctuate around +/- 24 FUEL.

For District Championship and/or *FIRST* Championship events, the quantity of FUEL staged in a MATCH may increase up to 600. If this happens, the additional FUEL will be included as part of an updated FUEL arrangement in the NEUTRAL ZONE.

6.3.4.1 NEUTRAL ZONE FUEL ARRANGEMENT

The NEUTRAL ZONE is staged by the FIELD crew by corralling all FUEL within a bounding box that is approximately 206.0in (5.23m) wide and 72.0in (1.83m) deep. The bounding box has a solid divider in the middle that is 2.0in (5.08cm) wide that will result in a small gap in FUEL. The dividers will be removed prior to the start of the MATCH.

Figure 6-3 NEUTRAL ZONE FUEL setup



The expected result of the dispersal is a roughly equal split of FUEL on both sides of the CENTER LINE. The dispersal between the scoring table side and non-scoring table side of the field is expected to vary match to match.

The placement of FUEL in the NEUTRAL ZONE is not intended to be in a perfect grid. Teams should expect variances in the placement of FUEL in the NEUTRAL ZONE at the start of the MATCH.

6.4 MATCH Periods

The first period of each MATCH is 20 seconds long and called the Autonomous Period (AUTO). During AUTO, ROBOTS operate without any DRIVE TEAM control or input. ROBOTS score SCORING ELEMENTS, may leave their ROBOT STARTING LINE and retrieve additional SCORING ELEMENTS, and may climb their TOWER. There is a 3-second delay between AUTO and TELEOP for scoring purposes as described in section [6.5 Scoring](#).

The second period of each MATCH is the remaining 2 minutes and 20 seconds (2:20) and called the Teleoperated Period (TELEOP). During TELEOP, DRIVERS remotely operate ROBOTS to retrieve and score SCORING ELEMENTS and climb their TOWER. TELEOP is split into further segments: the TRANSITION SHIFT, the ALLIANCE SHIFTS, and END GAME. There are four ALLIANCE SHIFTS named SHIFT 1, SHIFT 2, SHIFT 3, and SHIFT 4. The duration of each timeframe is shown in [Table 6-2](#).

Table 6-2: MATCH SHIFTS

MATCH Period	MATCH Timeframe	Duration	Timer Values
AUTO	AUTO	20 Seconds	0:20 – 0:00
TELEOP	TRANSITION SHIFT	10 Seconds	2:20 – 2:10
	SHIFT 1	25 Seconds	2:10 – 1:45
	SHIFT 2	25 Seconds	1:45 – 1:20
	SHIFT 3	25 Seconds	1:20 – 0:55
	SHIFT 4	25 Seconds	0:55 – 0:30
	END GAME	30 Seconds	0:30 – 0:00

6.4.1 HUB Status

During the MATCH, the status of a HUB can be either active or inactive. FUEL scored in an active HUB is worth MATCH points but FUEL scored in an inactive HUB will not earn any points as shown in [Table 6-3](#). Both ALLIANCE HUBS are active during AUTO, the TRANSITION SHIFT, and END GAME. During the ALLIANCE SHIFTS, only one ALLIANCE HUB will be active while the other ALLIANCE'S HUB becomes inactive.

The status of both HUBS during the ALLIANCE SHIFTS is based on the results of AUTO. The ALLIANCE that scores the most FUEL during AUTO will have their HUB set to inactive for SHIFT 1 while their opponent's HUB will be active, as shown in [Table 6-3](#). HUB statuses will then alternate at the start of each following ALLIANCE SHIFT, until the start of END GAME where both HUBS return to active. If both ALLIANCES score the same number of FUEL during AUTO, the FMS will randomly select an ALLIANCE and use its HUB status order for the ALLIANCE SHIFTS during the MATCH.

FMS Game Data relays the ALLIANCE who scored more FUEL during AUTO, or the ALLIANCE selected by FMS, to all OPERATOR CONSOLES simultaneously at the start of TELEOP. Lights on the HUB will also indicate the ALLIANCE HUB that will be inactive in ALLIANCE SHIFT 1 as noted in [Table 5-3](#). Specific details on the format of the data can be found on the [2026 FRC Control System website](#).

Table 6-3: Hub Status during MATCH Timeframes

AUTO Result:	RED ALLIANCE scores more FUEL during AUTO or is selected by the FMS		BLUE ALLIANCE scores more FUEL during AUTO or is selected by the FMS	
MATCH Timeframe (timer values)	RED ALLIANCE HUB status	BLUE ALLIANCE HUB status	RED ALLIANCE HUB status	BLUE ALLIANCE HUB status
AUTO (0:20 – 0:00)	Active	Active	Active	Active
TRANSITION SHIFT (2:20 – 2:10)	Active	Active	Active	Active
SHIFT 1 (2:10 – 1:45)	Inactive	Active	Active	Inactive
SHIFT 2 (1:45 – 1:20)	Active	Inactive	Inactive	Active
SHIFT 3 (1:20 – 0:55)	Inactive	Active	Active	Inactive
SHIFT 4 (0:55 – 0:30)	Active	Inactive	Inactive	Active
END GAME (0:30 – 0:00)	Active	Active	Active	Active

6.5 Scoring

ALLIANCES are rewarded for accomplishing various actions throughout a MATCH, including scoring FUEL, climbing their TOWER, and winning or tying MATCHES.

Rewards are granted either via MATCH points or Ranking Points (often abbreviated to RP, which increase the measure used to rank teams in the Qualification Tournament).

All scores are assessed and updated throughout the MATCH, except as follows:

- A. assessment of FUEL scored in the HUB continues for up to 3 seconds after the ARENA timer displays 0:00 following AUTO.
- B. assessment of FUEL scored in the HUB continues for up to 3 seconds after the ARENA timer displays 0:00 following TELEOP.
- C. assessment of AUTO TOWER points is made after the ARENA timer displays 0:00 following AUTO.

- D. assessment of TELEOP TOWER points is made 3 seconds after the ARENA timer displays 0:00 following TELEOP, or when all ROBOTS have come to rest following the conclusion of the MATCH, whichever happens first.

Assessment of FUEL scored in the HUB continues for 3 seconds after the HUB deactivates to account for FUEL processing time.

TOWER points are evaluated and scored by human volunteers. Teams are encouraged to make sure that it is obvious and unambiguous that the criteria are met.

6.5.1 SCORING ELEMENT Scoring Criteria

A FUEL is scored in the HUB once it passes through the top opening of the HUB and through the sensor array.

6.5.2 ROBOT Scoring Criteria

To qualify for TOWER points for a given LEVEL, a ROBOT must meet the following conditions:

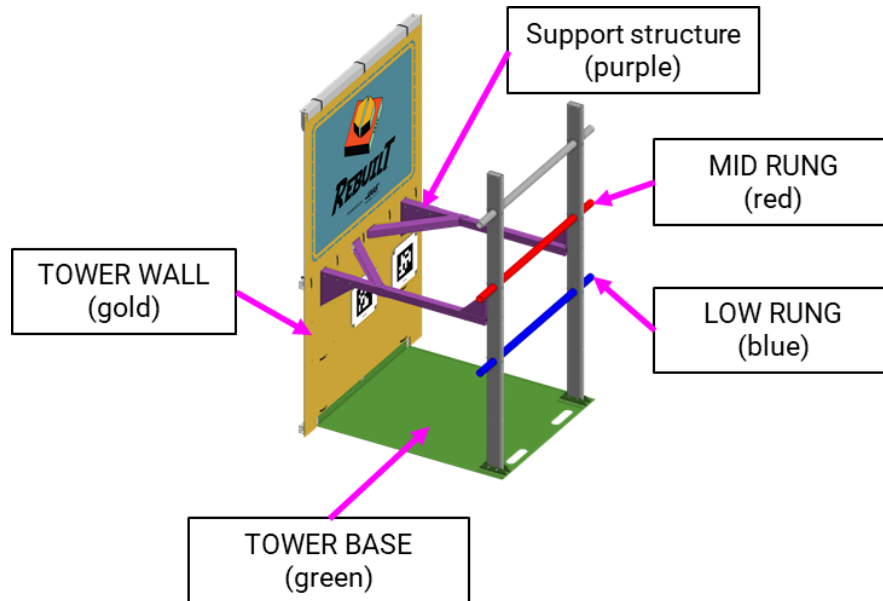
- For LEVEL 1 – a ROBOT must no longer be touching the carpet or the TOWER BASE, or
- For LEVEL 2 – a ROBOT must be positioned such that its BUMPER covers are completely above the LOW RUNG, or
- For LEVEL 3 – a ROBOT must be positioned such that its BUMPER covers are completely above the MID RUNG.

Additionally, a ROBOT **must be** contacting at least one RUNG and/or UPRIGHT on their TOWER and may additionally only contact the following elements:

- A. the TOWER WALL,
- B. support structure,
- C. FUEL, and/or.
- D. another ROBOT.

A ROBOT may only earn TOWER points for LEVEL 1 during AUTO. A ROBOT may only earn TOWER points for a single LEVEL during TELEOP. A ROBOT that earns TOWER points in AUTO is eligible to earn additional TOWER points during TELEOP.

Figure 6-4: TOWER contact limitation for ROBOT Scoring Criteria



6.5.3 Point Values

Point values for tasks in REBUILT are detailed in [Table 6-4](#).

Table 6-4 REBUILT point values

		MATCH points		Ranking Points
		AUTO	TELEOP	
FUEL	FUEL scored in an active HUB	1	1	
	FUEL scored in an inactive HUB	-	-	
TOWER	Each ROBOT at LEVEL 1 (2 ROBOTS max in AUTO)	15	10	
	Each ROBOT at LEVEL 2	-	20	
	Each ROBOT at LEVEL 3		30	
* ENERGIZED RP – The amount of FUEL scored in an active HUB is at or above threshold.				1
* SUPERCHARGED RP – The amount of FUEL scored in an active HUB is at or above threshold.				1
* TRAVERSAL RP – The amount of TOWER points scored during the MATCH is at or above threshold.				1
Win	completing a MATCH with more MATCH points than your opponent			3
Tie	completing a MATCH with the same number of MATCH points as your opponent			1

*See [Table 6-5](#) for threshold values. For District Championship and/or *FIRST* Championship events, the BONUS RP (ENERGIZED RP, SUPERCHARGED RP, and TRAVERSAL RP) requirement thresholds may increase.

Table 6-5: *REBUILT* BONUS RP thresholds

BONUS RP Type	Regional/ District Events	District Championships	<i>FIRST</i> Championship
ENERGIZED RP	100	TBA	TBA
SUPERCHARGED RP	360	TBA	TBA
TRAVERSAL RP	50	TBA	TBA

BONUS RP thresholds for District Championships and *FIRST* Championship will be announced in Team Updates.

6.6 Violations

Unless otherwise noted, all violations are assigned for each instance of a rule violation. A description of the penalties is listed in [Table 6-6](#). All rules throughout the Game Rules section are called as perceived by a REFEREE.

Table 6-6 Rule violations

Penalty	Description
MINOR FOUL	a credit of 5 points towards the opponent's MATCH point total
MAJOR FOUL	a credit of 15 points towards the opponent's MATCH point total
YELLOW CARD	issued by the Head REFEREE for egregious ROBOT or team member behavior or rule violations. A subsequent YELLOW CARD within the same tournament phase results in a RED CARD.
RED CARD	issued by the Head REFEREE for egregious ROBOT or team member behavior or rule violations which results in a team being DISQUALIFIED for the MATCH.
DISABLED	the state in which a ROBOT is commanded to deactivate all outputs, rendering the ROBOT inoperable for the remainder of the MATCH.
DISQUALIFIED	the state of a team in which they receive 0 MATCH points and 0 Ranking Points in a Qualification MATCH or causes their ALLIANCE to receive 0 MATCH points in a Playoff MATCH
VERBAL WARNING	a warning issued to a team by event staff or the Head REFEREE.
ALLIANCE is ineligible for RP	An ALLIANCE is ineligible for the specified RP for that MATCH. This overrides any RP awarded through normal MATCH play or other rule violations.

6.6.1 YELLOW and RED CARDS

In addition to rule violations explicitly listed throughout this document, YELLOW CARDS and RED CARDS are used in *FIRST* Robotics Competition to address team and ROBOT behavior that does not align with the mission, values, and culture of *FIRST*.

The Head REFEREE may assign a YELLOW CARD as a warning, or a RED CARD for egregious behavior inappropriate at a *FIRST* Robotics Competition event.

If an action resulting in the assignment of a YELLOW or RED CARD is determined to be the result of an ARENA FAULT, per section 10.2, the CARD will be rescinded.

A YELLOW or RED CARD is indicated on the audience display MATCH results screen. During Qualification MATCHES, A YELLOW or RED CARD is indicated next to the team who received the CARD and the Game Announcer describes the violation. During Playoff MATCHES, the card is applied to the whole ALLIANCE and as such "RED CARD" or "YELLOW CARD" appears above the ALLIANCE number.

YELLOW CARDS are additive, meaning that a second YELLOW CARD is automatically converted to a RED CARD. A team is issued a RED CARD for any subsequent incident in which they receive an additional YELLOW CARD, including earning a second YELLOW CARD during a single MATCH. A second YELLOW CARD is indicated with a RED CARD on the audience display next to the team who received the CARDS. A team that has received either a YELLOW CARD or a RED CARD carries a YELLOW CARD into subsequent MATCHES, except as noted below.

Once a team receives a YELLOW or RED CARD, a yellow rectangle will show next to the team number on the audience display during subsequent MATCHES, including any replays.

Figure 6-5 Example audience screen graphic showing YELLOW CARD indicators



All YELLOW CARDS are cleared in FMS at the conclusion of Practice, Qualification, and division Playoff MATCHES. VERBAL WARNINGS issued by the head REFEREE are cleared after Practice MATCHES and persist from Qualification MATCHES through subsequent tournament phases.

6.6.2 YELLOW and RED CARD application

YELLOW and RED CARDS are applied based on the following:

Table 6-7: YELLOW and RED CARD application

Time YELLOW or RED CARDS earned:	MATCH to which CARD is applied:
Prior to Qualification MATCHES	REFEREES may or may not be present during Practice MATCHES. With input from event staff, the Head REFEREE may opt to perpetuate a VERBAL WARNING or YELLOW CARD earned prior to Qualification MATCHES to the first Qualification MATCH for particularly egregious behavior.
during the Qualification MATCHES	team's current (or just-completed) MATCH. In the case where the team participated as a SURROGATE in the current (or just completed) MATCH, the card is applied to the team's previous MATCH (i.e. the team's second Qualification MATCH)
between the end of Qualification MATCHES and the start of Playoff MATCHES	ALLIANCE'S first Playoff MATCH
during the Playoff MATCHES	ALLIANCE'S current (or just-completed) MATCH

A MATCH is no longer the current MATCH once the results of the MATCH have been posted or the Head REFEREE or their designee has indicated that teams can collect their ROBOTS, whichever is later.

Please see examples of the application of YELLOW and RED CARDS as shown in section [6.6.4 Violation Details](#).

6.6.3 YELLOW and RED CARDS during Playoff MATCHES

During Playoff MATCHES, YELLOW and RED CARDS are assigned to the violating team's entire ALLIANCE instead of to only the violating team. If an ALLIANCE receives 2 YELLOW CARDS, the entire ALLIANCE is issued a RED CARD which results in DISQUALIFICATION for the associated MATCH. If both ALLIANCES receive a RED CARD, the ALLIANCE assessed the first RED CARD, chronologically, is DISQUALIFIED and loses the MATCH.

6.6.4 Violation Details

There are several styles of violation wording used in this manual. Below are some example violations and a clarification of the way the violation would be assessed. The examples shown do not represent all possible violations, but rather a representative set of combinations.

Table 6-8 Violation examples

Example Violation	Expanded Interpretation
MINOR FOUL	Upon violation, a MINOR FOUL is assessed against the violating ALLIANCE.
MAJOR FOUL and YELLOW CARD	Upon violation, a MAJOR FOUL is assessed against the violating ALLIANCE. After the MATCH, the Head REFEREE presents the violating team with a YELLOW CARD.
MINOR FOUL per additional SCORING ELEMENT. If egregious, YELLOW CARD	Upon violation, a number of MINOR FOULS are assessed against the violating ALLIANCE equal to the number of additional SCORING ELEMENTS beyond the permitted quantity. Additionally, if the REFEREES determine that the action was egregious, the Head REFEREE presents the violating team with a YELLOW CARD after the MATCH.
MINOR FOUL. MAJOR FOUL IF REPEATED	Upon initial violation in a MATCH, a MINOR FOUL is assessed against the violating ALLIANCE. If the ALLIANCE ROBOT repeats the infraction in the MATCH, a MAJOR FOUL is assessed against the violating ALLIANCE. Assuming no additional infractions of that rule by that ROBOT in that MATCH, the ROBOT has "earned" a MINOR FOUL and a MAJOR FOUL for their ALLIANCE.
MAJOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed	Upon violation, a MAJOR FOUL is assessed against the violating ALLIANCE and the REFEREE begins to count. Their count continues until the criteria to discontinue the count are met, and for each 3 seconds within that time,

Example Violation	Expanded Interpretation
	an additional MAJOR FOUL is assessed against the violating ALLIANCE. A ROBOT in violation of this type of rule for 15 seconds receives a total of 6 MAJOR FOULS (assuming no other rules were being simultaneously violated).
RED CARD for the ALLIANCE	After the MATCH, the Head REFEREE presents the violating ALLIANCE with a RED CARD in the following fashion: • In a PLAYOFF MATCH, a single RED CARD is assessed to the ALLIANCE. • In all other scenarios, each team on the ALLIANCE is issued a RED CARD.

6.7 Head REFEREE and FTA Interaction

The Head REFEREE has the ultimate authority in the ARENA during the event, but may receive input from additional sources, e.g. Game Designers, *FIRST* personnel, FTA, and other event staff. The Head REFEREE rulings are final. No event staff, including the Head REFEREE, will review video, photos, artistic renderings, etc. of any MATCH, from any source, under any circumstances.

6.7.1 Question Box (Q)

Each ALLIANCE has a designated Question Box near the scoring table. If a DRIVE TEAM has a question about a MATCH, the FIELD, etc., they may send up to 2 DRIVE TEAM members to their corresponding Question Box. Depending on timing, the Head REFEREE or FTA may postpone any requested discussion until the end of the subsequent MATCH.

Technical questions regarding FIELD or ROBOT operation are addressed by the FTA, and additional team members are invited to participate in these conversations if necessary. If a DRIVE TEAM needs clarification on a ruling or score, per [Q101](#), up to 2 DRIVE TEAM members should address the Head REFEREE after the ARENA Reset Signal (e.g. FIELD lights turn green).

While FMS tracks quantities of MINOR and MAJOR FOULS, *FIRST* instructs REFEREES to not self-track details about MINOR FOULS and MAJOR FOULS; as a result, REFEREES are not expected to recall details about what MINOR FOULS and MAJOR FOULS were made, when they occurred, and against whom.

Any reasonable question is fair game in the Question Box, and Head REFEREES will make good faith efforts to provide helpful feedback (e.g. how/why certain MINOR FOULS are being called, why a particular ROBOT may be susceptible to certain MINOR FOULS based on its design or game play, how specific rules are being called or interpreted), but please know that they may not be able to supply specific details.

Q101 *Head REFEREE Interactions. A team may only address the Head REFEREE with maximum of 2 people, 1 of which must be a STUDENT.

Violation: The Head REFEREE will not address non-compliant team members or peripheral conversations.

Some events may restrict ARENA access to members of the DRIVE TEAM. Team members are permitted to swap buttons within their team as needed to access the Question Box.

As much as possible, the STUDENT should be an active participant in the conversation.

Teams should not record interactions without consent (see [E117](#)).

Q102 Conversations must be civil. Any team interaction with the Head REFEREE must be appropriate.

Violation: The Head REFEREE may terminate the conversation to allow time for parties to cool down. A person with subsequent violations will not be allowed to interact with the Head REFEREE.

Examples of inappropriate behavior are outlined in [G201](#).

6.8 Other Logistics

SCORING ELEMENTS that leave the FIELD (other than through the opening at the base of the OUTPOST) are placed back into the FIELD approximately at the point of exit by FIELD STAFF (REFEREES, FTAs, or other staff working around the FIELD) at the earliest safe opportunity.

Note that ROBOTS and HUMAN PLAYERS may not deliberately cause SCORING ELEMENTS to leave the FIELD (see [G405](#)).

An ARENA FAULT (an error in ARENA operation described in section [10.2 MATCH Replays](#)) is not called for MATCHES that accidentally begin with damaged SCORING ELEMENTS. Damaged SCORING ELEMENTS are not replaced until the next ARENA reset period. DRIVE TEAMS should alert the FIELD STAFF to any missing or damaged SCORING ELEMENTS prior to the start of the MATCH but as noted in [6.3.4](#), the exact number of FUEL in the NEUTRAL ZONE may vary.

Once the MATCH is over and the Head REFEREE determines that the FIELD is safe for FIELD STAFF and DRIVE TEAMS, they or their designee change the FIELD lights to green and DRIVE TEAMS may retrieve their ROBOT. Some MATCHES may include a short period afterward to allow FIELD staff to begin to clean the field to make it easier and safer for teams to collect their robot. This period is indicated with purple lights on the team signs and the word “clean” displayed. TEAMS are not allowed on the field during this time.

During ARENA reset, the ARENA is cleared of ROBOTS and OPERATOR CONSOLES from the MATCH that just ended, ROBOTS and OPERATOR CONSOLES for the subsequent MATCH are loaded into the ARENA by DRIVE TEAMS, and FIELD STAFF reset ARENA elements.

FIRST Robotics Competition uses 3 words in the context of how durations and actions are assessed with regards to evaluation of rules and assignment of violations. These words provide general guidance to describe benchmarks to be used across the program. It is not the intent for REFEREES to provide a count during the time periods.

- MOMENTARY describes durations that are fewer than approximately 3 seconds.
- CONTINUOUS describes durations that are more than approximately 10 seconds.
- REPEATED describes actions that happen more than once within a MATCH.



7 Game Rules (G)

7.1 Personal Safety

G101 ***Humans, remain outside the FIELD.** A team member may not reach into the FIELD with any part of their body during a MATCH .

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

Violations of this rule apply to the entire team, not specifically to any 1 individual. For example, a member of Team 9999 reaches onto the FIELD during MATCH 3, and a different member reaches onto the FIELD during MATCH 25. The team receives a VERBAL WARNING for the first violation and a YELLOW CARD for the second.

G102 ***Never step over the guardrail.** A team member may only enter or exit the FIELD through open gates and only enter if FIELD lighting (FIELD facing side of the team signs and timers) is green, unless explicitly instructed by a REFEREE or an FTA.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

Teams are encouraged to ensure that all members of their DRIVE TEAM are aware of this rule. It's easy to violate, particularly when teams are doing their best to move on and off the FIELD quickly. The violations of this rule are intended to avoid nuisance penalties, but still enforce safety requirements around the FIELD. There is the potential for injury when stepping over the guardrail.

Violations of this rule apply to the entire team, not specifically to any 1 individual. For example, a member of Team 9999 steps over the guardrail prior to MATCH 3, and a different member steps over the guardrail prior to MATCH 25. The team receives a VERBAL WARNING for the first violation and a YELLOW CARD for the second.

G103 ***Be careful what you interact with.** A team member is prohibited from the following actions with regards to interaction with ARENA elements.

- A. climbing on or inside,
- B. hanging from,
- C. manipulating such that it doesn't return to its original shape without human intervention, and
- D. damaging.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

The REBUILT FIELD has obstacles, and caution should be applied when navigating the FIELD. Teams should avoid carrying their ROBOT over the BUMP. Walking over the BUMP is not considered climbing on.

- G104** ***Teams may not enable their ROBOTS on the FIELD.** Teams may not tether to the ROBOT while on the FIELD except in special circumstances (e.g. after Opening Ceremonies, before an immediate MATCH replay, etc.) and with the express permission from the FTA or a REFEREE.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

Teams are encouraged to consider this rule when developing their ROBOTS.

FMS will not enable ROBOTS after the conclusion of the MATCH.

For the purposes of this rule, tethering includes any wired or wireless connection used to electrically energize and/or control elements on the ROBOT. The safety of teams and volunteers in close proximity to ROBOTS and ARENA elements on the FIELD is of the utmost importance, therefore ROBOTS or ROBOT COMPONENTS may not be enabled in any way on the FIELD before or after the MATCH.

ROBOTS need to be safely transported off the FIELD and back to the pits after the MATCH, and there may be bystanders, doorways, or height restrictions along the route.

7.2 Conduct

- G201** ***Be a good person.** All teams must be civil toward everyone and respectful of team and event equipment while at a FIRST Robotics Competition event.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

Examples of inappropriate behavior include, but are not limited to, the use of offensive language or other uncivil conduct.

Examples of particularly contemptible behavior that is likely to result in ARENA ejection include, but are not limited to, the following:

- A. assault, e.g. throwing something that hits another person (even if unintended),
- B. threat, e.g. saying something like “if you don’t reverse that call, I’ll make you regret it,”
- C. harassment, e.g. badgering someone with no new information after a decision’s been made or a question’s been answered,
- D. bullying, e.g. using body or verbal language to cause another person to feel inadequate,
- E. insulting, e.g. telling someone they don’t deserve to be on a DRIVE TEAM,
- F. swearing at another person (versus swearing under one’s breath or at oneself), and
- G. yelling at another person(s) in anger or frustration.

- G202** ***Don’t bang on the glass.** A team member may never strike or hit the DRIVER STATION plastic windows.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

G203 ***Asking other teams to throw a MATCH – not cool.** A team may not encourage an ALLIANCE of which it is not a member to play beneath its ability.

Note: This rule is not intended to prevent an ALLIANCE from planning and/or executing its own strategy in a specific MATCH in which all the teams are members of the ALLIANCE.

Violation: *VERBAL WARNING. RED CARD if subsequent violations during the event.*

Example 1: A MATCH is being played by Teams A, B, and C, in which Team C is encouraged by Team D to not get on the TOWER at the end of the MATCH, resulting in Teams A, B, and C not earning a Ranking Point. Team D's motivation for this behavior is to prevent Team A from rising in the Tournament rankings and negatively affecting Team D's ranking. Team D has violated this rule.

Example 2: A MATCH is being played by Teams A, B, and C, in which Team A is assigned to participate as a SURROGATE. Team D encourages Team A to not participate in the MATCH so that Team D gains ranking position over Teams B and C. Team D has violated this rule.

Example 3: Asking a team to "no show" for a MATCH.

FIRST considers the action of a team influencing another team to throw a MATCH, to deliberately miss Ranking Points, etc. incompatible with FIRST values and not a strategy any team should employ.

G204 ***Letting someone coerce you into throwing a MATCH – also not cool.** A team, as the result of encouragement by a team not on their ALLIANCE, may not play beneath its ability.

Note: This rule is not intended to prevent an ALLIANCE from planning and/or executing its own strategy in a specific MATCH in which all the ALLIANCE members are participants.

Violation: *VERBAL WARNING. RED CARD if subsequent violations during the event.*

Example 1: A MATCH is being played by Teams A, B, and C. Team D requests Team C ignore the TOWER at the end of the MATCH, resulting in Teams A, B, and C not being able to earn the TRAVERSAL RP. Team C accepts this request from Team D. Team D's motivation for this behavior is to prevent Team A from rising in the Tournament rankings negatively affecting Team D's ranking. Team C has violated this rule.

Example 2: A MATCH is being played by Teams A, B, and C, in which Team A is assigned to participate as a SURROGATE. Team A accepts Team D's request not to participate in the MATCH so that Team D gains ranking position over Teams B and C. Team A has violated this rule.

FIRST considers the action of a team influencing another team to throw a MATCH, to deliberately miss Ranking Points, etc. incompatible with FIRST values and not a strategy any team should employ.

- G205 *Throwing your own MATCH is bad.** A team may not intentionally lose a MATCH or sacrifice Ranking Points in an effort to lower their own ranking or manipulate the rankings of other teams.

Violation: VERBAL WARNING. RED CARD if subsequent violations during the event.

The intent of this rule is not to punish teams who are employing alternate strategies, but rather to ensure that it is clear that throwing MATCHES to negatively affect your own rankings, or to manipulate the rankings of other teams (i.e. throw a MATCH to lower a partner's ranking, and/or increase the ranking of another team not in the MATCH) is incompatible with *FIRST* values and not a strategy any team should employ.

- G206 *Don't violate rules for RPs.** A team or ALLIANCE may not collude with another team to each purposefully violate a rule in an attempt to influence Ranking Points.

Violation: YELLOW CARD and the ALLIANCE is ineligible for the BONUS RPs.

For example, if Team A on the blue ALLIANCE agrees with Team F on the red ALLIANCE to contact each other while each ROBOT is climbing the TOWER resulting in both ALLIANCES being awarded the TRAVERSAL RP.

- G207 *Don't abuse ARENA access.** A team member (except DRIVERS, HUMAN PLAYERS, and DRIVE COACHES) granted access to restricted areas in and around the ARENA (e.g. via TECHNICIAN button, event issued Media badges, etc.) may not assist or use signaling devices during the MATCH. Exceptions will be granted for inconsequential infractions and in cases concerning safety.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations during the event.

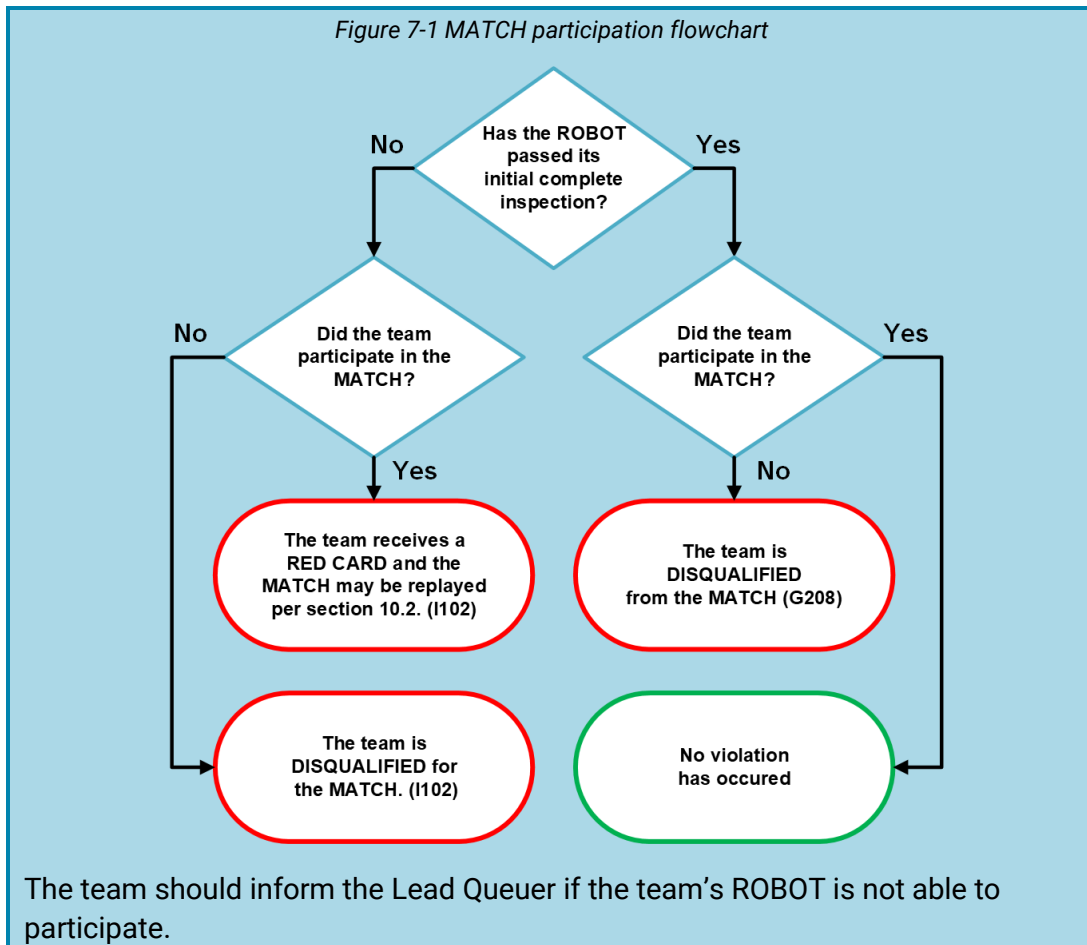
The TECHNICIAN'S role is to help the team prepare the ROBOT so it can perform at its full potential during a MATCH. The TECHNICIAN, except as described at the end of section [6.2 DRIVE TEAM](#), is not an additional DRIVE COACH, DRIVER, or HUMAN PLAYER.

Team members in open-access spectator seating areas are not considered to be in a restricted area and are not prevented from assisting or using signaling devices. See [E102](#) for related details.

- G208 *Show up to your Qualification MATCHES.** If a ROBOT has passed initial, complete inspection, at least 1 member of its DRIVE TEAM must report to the ARENA and participate in each of their assigned Qualification MATCHES.

Violation: DISQUALIFIED from the current MATCH.

Figure 7-1 MATCH participation flowchart



G209 ***Keep your ROBOT together.** A ROBOT may not intentionally detach or leave a part on the FIELD.

Violation: *RED CARD.*

G210 ***Don't expect to gain by doing others harm.** A strategy not consistent with standard gameplay and clearly aimed at forcing the opponent ALLIANCE to violate a rule is not in the spirit of FIRST Robotics Competition and not allowed. Rule violations forced in this manner will not result in an assignment of a penalty to the targeted ALLIANCE.

Violation: *MAJOR FOUL. YELLOW CARD if REPEATED.*

This rule does not apply for strategies consistent with standard gameplay, for example:

- A. a red ALLIANCE ROBOT attempting to climb their TOWER in the final 30 seconds of the MATCH contacts a blue ALLIANCE ROBOT.
- B. a blue ALLIANCE ROBOT is attempting to play defense on a red ALLIANCE ROBOT attempting to score into the HUB from within their ALLIANCE ZONE and pushes them outside their ALLIANCE ZONE.

This rule requires an intentional act with limited or no opportunity for the team being acted on to avoid the penalty such as:

- C. forcing an opponent ROBOT to catch FUEL by pushing them far from (i.e. more than 48.0in (1.22m)) the HUB exit to under the HUB exit to make them violate [G408](#).
- D. a blue ALLIANCE ROBOT, pushing a red ALLIANCE ROBOT far from (i.e. more than 48.0in (1.22m)) the TOWER into another blue ALLIANCE ROBOT which is in contact with the blue TOWER and the REFEREE perceives that the blue ROBOT is deliberately making the red ROBOT violate [G420](#).

G211 *Egregious or exceptional violations. Egregious behavior beyond what is listed in the rules or subsequent violations of any rule or procedure during the event is prohibited.

In addition to rule violations explicitly listed in this manual and witnessed by a REFEREE, the Head REFEREE may assign a YELLOW or RED CARD for egregious ROBOT actions or team member behavior at any time during the event.

Please see section [6.6.1 YELLOW and RED CARDS](#) for additional detail.

Violation: YELLOW or RED CARD.

The intent of this rule is to provide the Head REFEREES the flexibility necessary to keep the event running smoothly, as well as keep the safety of all the participants as the highest priority. Behaviors that put the *FIRST* community or integrity of the game at risk are not allowed and are violations of this rule. Those behaviors include, but are not limited to the list below:

- A. inappropriate behavior as outlined in the blue box of [G201](#),
- B. jumping over the guardrail,
- C. pushing past the FIELD reset person blocking an open gate to get on the FIELD,
- D. reaching into the FIELD and grabbing a ROBOT during a MATCH,
- E. a single PIN in excess of 15 seconds,
- F. exploiting the 3-second window after a MATCH described in section [6.5 Scoring](#) to avoid rule violations (e.g. triggering an over-extension that enables TOWER points or using a ROBOT'S residual energy to impact an opponent ROBOT on their TOWER),
- G. triggering scoring sensors or otherwise interfering with FMS or FIELD operation,
- H. intentionally crossing the CENTER LINE in AUTO and contacting an opponent ROBOT in order to interfere with an opponent ROBOT'S AUTO,
- I. intentionally ejecting a large quantity of FUEL out of the FIELD,
- J. intentionally exceeding the expansion limits for strategic benefit (i.e. climb the TOWER, block part of the FIELD, etc.),
- K. a HUMAN PLAYER hoarding a large quantity of FUEL outside the designated locations in [G427](#),
- L. Intentionally scoring a large quantity of FUEL into the HUB from the NEUTRAL ZONE.

The Head REFEREE may assign a YELLOW or RED CARD for a single instance of a rule violation such as the examples given in items above, or for multiple

instances of any single rule violation. Teams should be aware that any rule in this manual could escalate to a YELLOW or RED CARD. The Head REFEREE has final authority on all rules and violations at an event.

G212 *All teams can play. A team may not encourage another team to exclude their ROBOT or be BYPASSED from a qualification MATCH for any reason.

Violation: YELLOW CARD. RED CARD if the ROBOT does not participate in the MATCH.

7.3 Pre-MATCH

G301 *Be prompt. A DRIVE TEAM member may not cause significant delays to the start of their MATCH. Causing a significant delay requires both of the following to be true:

A. The expected MATCH start time has passed, and

Event volunteers communicate schedule delays with teams to the best of their ability. The Pit Display (which is typically located near the Pit Administration desk) shows any event timing delay. Announcements on the FIELD and in the pits also provide information on delays, and any team uncertain of when to queue for a MATCH should communicate with queuing volunteers.

During Qualification MATCHES, the expected start time of the MATCH is the time indicated on the MATCH schedule or the cycle time for current round minus 3 minutes from the end of the previous MATCH (which is reflected on the schedule on the Pit Display), whichever is later.

During Playoff MATCHES, the expected start time of the MATCH is the time indicated on the MATCH schedule or 15 minutes from either ALLIANCE'S previous MATCH, whichever is later.

B. The DRIVE TEAM has access to the FIELD and is neither MATCH ready nor making a good faith effort, to quickly become MATCH ready.

Teams that have violated [G208](#) or have 1 DRIVE TEAM member present and have informed event staff that their ROBOT will not be participating in the MATCH are considered MATCH ready and not in violation of this rule.

Violation: If a Qualification MATCH, VERBAL WARNING. MAJOR FOUL for the upcoming MATCH if a subsequent violation occurs within the tournament phase. If the DRIVE TEAM is not MATCH ready within 2 minutes of the VERBAL WARNING/MAJOR FOUL and the Head REFEREE perceives no good faith effort by the DRIVE TEAM to quickly become MATCH ready, DISABLED.

If a Playoff MATCH, VERBAL WARNING issued to the ALLIANCE. MAJOR FOUL for the ALLIANCE'S upcoming MATCH if a subsequent violation occurs within the tournament phase. If the ALLIANCE is not MATCH ready within 2 minutes of the VERBAL WARNING/MAJOR FOUL having been issued to all 3 teams and the Head REFEREE perceives no good faith effort by the DRIVE TEAM(s) to quickly become MATCH ready, the offending team's ROBOT is DISABLED.

The intent of this rule is to provide an equitable amount of time for both ALLIANCES to prepare for each MATCH and give DRIVE TEAMS grace given extenuating circumstances that cause them to be late.

Once a VERBAL WARNING /MAJOR FOUL is issued, the Head REFEREE starts a 2-minute timer and makes a good faith effort to share the timer's status with the delaying DRIVE TEAM.

Being "MATCH ready" requires that the ROBOT is on the FIELD, in its STARTING CONFIGURATION, and turned on. Additionally, the DRIVE TEAM members must be in their starting positions.

In general, good faith efforts to quickly become MATCH ready are entirely for the purposes of transitioning the ROBOT into a MATCH ready state (i.e. not attempts to significantly alter a ROBOT's capabilities.) Examples of good faith efforts to quickly become MATCH ready include but are not limited to:

- A. walking safely towards the FIELD with a ROBOT that a team is not actively modifying,
- B. applying quick fixes such as tape or cable ties to make the ROBOT compliant with STARTING CONFIGURATION requirements,
- C. waiting for an OPERATOR CONSOLE computer to boot, and
- D. working with FIELD STAFF to get the ROBOT connected to the FIELD.

Examples that are not considered good faith efforts to quickly become MATCH ready include but are not limited to:

- E. a ROBOT not moving to the FIELD,
- F. a ROBOT moving to the FIELD but being actively modified while doing so,
- G. a DRIVE TEAM member remaining on the FIELD once a MATCH is ready to begin (indicated by the green LEDs having turned off),
- H. installing BUMPERS, charging pneumatic systems, or any other ROBOT maintenance not considered a quick fix as described in item B above once on the FIELD,
- I. time-consuming use of alignment devices that are external to the ROBOT (e.g. a DRIVE TEAM could bring and use a measuring tape, as long as there is no delay to the MATCH by doing so), and
- J. repairs more substantial than those described in B.

There are no rules that prohibit use of hand tools (including battery operated tools) while setting up ROBOTS from the FIELD, provided they do not cause significant delay or cause safety concerns.

G302 *Limit what you use during a MATCH. Items used during a match must fit on your team's DRIVER STATION shelf, be worn or held by members from your DRIVE TEAM, or be an item used as an accommodation (e.g. stools, crutches, etc.). Regardless of if the equipment fits the criteria above, it may not:

- A. be employed in a way that introduces a safety hazard,
- B. extend more than 78.0in (1.981m) above the floor,
- C. communicate with anything or anyone outside of the ARENA with the exception of medically required equipment,
- D. block visibility for FIELD STAFF or audience members, or
- E. jam or interfere with the remote sensing capabilities of another team or the FIELD.

Exceptions to part B are granted for momentary extensions above 78.0in (1.91m) and for individuals above 78.0in (1.91m) wearing PPE or reasonable decorative apparel such as hats, headbands, etc.

Violation: MATCH will not start until the situation is remedied. If discovered or used inappropriately during a MATCH, YELLOW CARD.

Examples of equipment that may be considered a safety hazard in the confined space of the ALLIANCE AREA include, but are not limited to, a folding step stool, ladder, or a large signaling device.

Using an item that has wireless communications disabled complies with C above.

Examples of jamming or interfering with remote sensing capabilities include, but are not limited to, mimicking the FIELD AprilTags and shining bright lighting or laser pointers onto the FIELD.

G303 *Start your ROBOTS. A ROBOT must meet all following MATCH-start requirements:

- A. it does not pose a hazard to humans, FIELD elements, or other ROBOTS,
- B. has passed initial, complete inspection, i.e. it's compliant with all ROBOT rules (for exceptions regarding Practice MATCHES, see section [9 Inspection & Eligibility \(I\)](#)),
- C. if modified after initial Inspection, it's compliant with [I104](#),
- D. its BUMPERS overlap their ROBOT STARTING LINE,
- E. it's not contacting the BUMP,
- F. it's the only team-provided item left on the FIELD,
- G. it's not attached to, entangled with, or suspended from any FIELD element,
- H. it's confined to its STARTING CONFIGURATION (reference [R102](#) and [R104](#)), and
- I. it fully and solely supports not more than 8 FUEL (as described in section [6.3.4 SCORING ELEMENTS](#)).

Violation: If fix is a quick remedy, the MATCH won't start until all requirements are met. If it is not a quick remedy, DISABLED and, at the discretion of the Head REFEREE, must be re-inspected. If a ROBOT not compliant with part B or C participates, its team receives a RED CARD.

If a ROBOT is BYPASSED prior to the start of the MATCH, the DRIVE TEAM may not remove the ROBOT from the FIELD without permission from the Head REFEREE or the FTA.

For assessment of many of the items listed above, the Head REFEREE is likely to consult with the LRI.

ROBOTS may be asked to move if at least 1 of the FIELD gates for each ALLIANCE are not able to open/close when the ROBOT is in its starting location. Teams are encouraged to have multiple AUTO starting locations.

Locations for FIELD gates can be found in the [Field Dimension Drawings](#).

7.4 In-MATCH

Rules in this section pertain to game play once a MATCH begins.

7.4.1 AUTO

The AUTO period is the first 20 seconds of the MATCH, and the FMS blocks any DRIVER control, so ROBOTS operate with only their pre-programmed instructions. Rules in this section apply only during the AUTO period.

G401 *Behind the lines. In AUTO, each DRIVE TEAM member must remain in their staged areas. A DRIVE TEAM member staged behind a HUMAN STARTING LINE may not contact anything in front of that HUMAN STARTING LINE, unless for personal or equipment safety, to press the E-Stop or A-Stop, or granted permission by a Head REFEREE or FTA.

Violation: MINOR FOUL regardless of the number of items contacted.

Pointing, gesturing, or otherwise extending across the HUMAN STARTING LINE such that contact is not made with carpet or other ARENA elements is not a violation of this rule.

An example of an exception for equipment safety is if an OPERATOR CONSOLE starts to fall from, or has already fallen off of, the DRIVER STATION shelf. In that circumstance, DRIVE TEAM members may step forward to catch it or pick it up off the ground and return it to the shelf.

G402 *Let the ROBOT do its thing. In AUTO, a DRIVE TEAM member may not directly or indirectly interact with a ROBOT or an OPERATOR CONSOLE unless for personal safety, OPERATOR CONSOLE safety, or pressing an E-Stop or A-Stop. A HUMAN PLAYER entering FUEL onto the FIELD is an exception to this rule.

Violation: MINOR FOUL and YELLOW CARD.

G403 Limited AUTO opponent interaction. In AUTO, a ROBOT whose BUMPERS are completely across the CENTER LINE (i.e. to the opposite side of the CENTER LINE from its ROBOT STARTING LINE) may not contact an opponent ROBOT.

Violation: MAJOR FOUL.

7.4.2 SCORING ELEMENTS

G404 *ROBOTS: use SCORING ELEMENTS as directed. A ROBOT may not deliberately use a SCORING ELEMENT in an attempt to ease or amplify a challenge associated with a FIELD element.

Violation: MAJOR FOUL.

Examples include, but are not limited to:

- A. launching SCORING ELEMENTS at opponent ROBOTS,
- B. using SCORING ELEMENTS to elevate ROBOTS in an attempt to climb the TOWER,
- C. positioning SCORING ELEMENTS to impede opponent access to their TOWER,

G405 *Keep SCORING ELEMENTS in bounds. A ROBOT may not intentionally eject SCORING ELEMENTS from the FIELD (either directly or by bouncing off a FIELD element or other ROBOT) with an exception of through the opening at the base of the OUTPOST.

Violation: MINOR FOUL. If REPEATED, MAJOR FOUL.

G406 *Don't abuse SCORING ELEMENTS. Neither a ROBOT nor a HUMAN PLAYER may damage a SCORING ELEMENT.

Violation: VERBAL WARNING. MAJOR FOUL if REPEATED in any subsequent MATCHES during the event. If via a ROBOT and the Head REFEREE determines that further damage is likely to occur, DISABLED. Corrective action (such as eliminating sharp edges, removing the damaging MECHANISM, and/or re-inspection) may be required before the ROBOT may compete in subsequent MATCHES.

SCORING ELEMENTS are expected to undergo a reasonable amount of wear and tear as they are handled by ROBOTS, such as scratching or marking. Routinely gouging, tearing off pieces, or marking SCORING ELEMENTS are violations of this rule.

G407 Only score while in your ALLIANCE ZONE. A ROBOT may not launch a SCORING ELEMENT into their HUB unless their BUMPERS are partially or fully within their ALLIANCE ZONE.

Violation: MAJOR FOUL

G408 Don't catch FUEL. A ROBOT may not do either of the following with FUEL released by the HUB unless and until that FUEL contacts anything else besides that ROBOT or FUEL CONTROLLED by that ROBOT:

- A. gain greater than MOMENTARY CONTROL of FUEL, or
- B. push or redirect FUEL to a desired location or in a preferred direction.

A ROBOT is in CONTROL of a SCORING ELEMENT if the SCORING ELEMENT is fully supported by or stuck in, on, or under the ROBOT.

Violation: MINOR FOUL. If strategic, MAJOR FOUL and VERBAL WARNING. If subsequent strategic violations during the event, MAJOR FOUL and YELLOW CARD.

Examples of interaction with a SCORING ELEMENT that are not "CONTROL" include, but are not limited to:

- A. "bulldozing" (inadvertent contact with a SCORING ELEMENT while in the path of the ROBOT moving about the FIELD),
- B. "deflecting" (being hit by a SCORING ELEMENT that bounces off a ROBOT and moves in a random direction.)

Examples of interaction which would be considered strategic include, but are not limited to:

- C. intentionally sitting under the HUB to collect a large quantity of FUEL,
- D. intentionally sitting under the HUB in order to redirect FUEL into your ALLIANCE ZONE.

Violations of this rule are per instance and not per FUEL CONTROLLED. Generally, an instance is considered for each time a ROBOT is catching or redirecting a quantity of FUEL from the HUB. Any time flow of FUEL into the robot has stopped would be considered a separate instance. Generally, any action which attempts to exploit the definition of an instance of any rule in order to gain benefit will likely be a violation of [G211](#) and would quickly escalate to a RED CARD.

7.4.3 ROBOT

G409 *ROBOTS must be safe. A ROBOT may not pose an undue hazard to a human, an ARENA element, or another ROBOT in the following ways:

- A. the ROBOT or anything it CONTROLS, e.g. FUEL, contacts anything outside the FIELD except for MOMENTARY contact inside the CHUTE and/or the CORRAL,
- B. its BUMPERS fail such that a segment completely detaches,
- C. a corner of its ROBOT PERIMETER is exposed,
- D. its team number or ALLIANCE color are indeterminate,
- E. its BUMPERS leave the BUMPER ZONE (see R405) REPEATEDLY or for more than a MOMENTARY amount of time, or
- F. its operation or design is dangerous or unsafe.

Violation: DISABLED. Corrective action (such as repair of BUMPERS, removing the unsafe MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.

Examples of dangerous operation or designs that likely pose undue hazards include, but are not limited to:

- A. uncontrolled motion that cannot be stopped by the DRIVE TEAM,
- B. ROBOT parts “flailing” outside of the FIELD,
- C. ROBOTS dragging their battery, and
- D. ROBOTS that consistently extend outside the FIELD.

Please be conscious of REFEREES and FIELD STAFF working around the ARENA who may be in close proximity to your ROBOT.

G410 *Keep your BUMPERS low. ROBOT extensions may not interact with the carpet, BUMPS, or TOWER BASE such that the BUMPERS are lifted out of the BUMPER ZONE (see [R405](#)).

Violation: MINOR FOUL. Corrective action (such as removing the offending MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.

G411 Don't damage the FIELD. A ROBOT may not damage FIELD elements.

Violation: VERBAL WARNING. If the Head REFEREE infers that additional damage is likely, DISABLED. YELLOW CARD for any subsequent damage during the event.

Corrective action (such as eliminating sharp edges, removing the damaging MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.

G412 Watch your FIELD interaction. A ROBOT is prohibited from the following interactions with FIELD elements (with the exception of the RUNGS and UPRIGHTS):

- A. grabbing,
- B. grasping,
- C. attaching to (including the use of a vacuum or hook fastener to anchor to the FIELD carpet),
- D. becoming entangled with, and
- E. suspending from.

Violation: MAJOR FOUL, plus YELLOW CARD if REPEATED, or longer than MOMENTARY. If the Head REFEREE infers that damage is likely, DISABLED. Corrective action (such as removing the offending MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.

G413 Expansion limits. A ROBOT may not extend beyond any of the horizontal or vertical expansion limits described in [R105](#), [R106](#), and [R107](#).

Exceptions to this rule:

- A. If the over-expansion that violates [R105](#) or [R107](#) is due to visible damage and not used for strategic benefit, no penalty is imposed.
- B. If an expansion that contributes to a violation of [R106](#) is due to visible damage the team may extend a different component in a different direction, and no penalty is imposed.
- C. If the over-expansion is MOMENTARY and is not used for strategic benefit, no penalty is imposed.
- D. If an expansion that contributes to a violation of [R106](#) is due to flex of ROBOT components and is within 1.5in of the ROBOT PERIMETER, the team may extend in a different direction, and no penalty is imposed.

Violation: MAJOR FOUL. Corrective action (such as removing the offending MECHANISM, and/or re-inspection) may be required before the ROBOT will be allowed to compete in subsequent MATCHES.

The intent of the [G413-A](#) and [G413-B](#) exceptions to this rule are to prevent piling on a punitive response to a ROBOT that's already experienced hardship and not leveraging that hardship for gain.

Exceptions are only given for visible damage, as perceived by a REFEREE. Teams should not assume that REFEREES will give an exception for unobservable damage even if ROBOT function is affected.

Teams exploiting the exception to part B by designing in something to "break" will not be given an exception and will likely also be given a violation of [G211](#).

Examples related to exceptions A and B include the following:

- A. a physical device on a team's ROBOT, whose purpose is to restrain their TOWER mechanism from extending beyond the 30in (76.2cm) height limit imposed by [R107](#), breaks after a collision with another ROBOT. Provided the

ROBOT does not use the now-too-long extension to climb the TOWER, no violation is assigned.

- B. a vertical structural member of a ROBOT breaks at the bottom and rotates out such that it exceeds the 12in (30.48cm) limit imposed by [R105](#). The ROBOT then parks such that its extension blocks opponent ROBOTS from reaching the OUTPOST. A MAJOR FOUL is issued.
- C. a part of a ROBOT is damaged causing a panel to extend out less than 12in (30.48cm) on one side of the ROBOT. The ROBOT then extends out in another direction to intake FUEL. As visible damage has caused an expansion that contributes to the violation of [R106](#), no penalty is imposed.
- D. a mechanism that controls a ROBOT'S intake is damaged in a way that's not visible to a REFEREE and the team can no longer bring their intake back in. The team then extends out in another direction to climb the TOWER. The intake is not visibly damaged, so a violation of a MAJOR FOUL is issued.

The intent of the [G413-C](#) extension is to prevent violations for inadvertent mechanism flexing and movement for short periods of time during the MATCH not to provide a pathway to intentionally design less-than-MOMENTARY over-extensions.

Examples related to exception C include the following:

- E. A ROBOT has a mechanism deployed out over one side of the ROBOT PERIMETER which, due to ROBOT movement or a collision, flexes MOMENTARILY beyond the projection of that side of the ROBOT PERIMETER. As the action is less than MOMENTARY and not used for strategic benefit, no penalty is imposed.
- F. A ROBOT has a mechanism designed to rotate from side A to side B of their ROBOT and MOMENTARILY extends out both side A and B during rotation. This action is used for strategic benefit so a violation of a MAJOR FOUL is issued.

For example F, the ROBOT will likely need to take corrective action before being allowed to compete in subsequent MATCHES.

Note that [G211](#) may apply if a team is intentionally exceeding the expansion limits for strategic benefit.

At the conclusion of the MATCH, the Head REFEREE may elect to visually inspect a ROBOT and remove the violation if the damage is verified.

G414 Don't Climb on each other. ROBOTS may not fully support the weight of other ROBOTS on their ALLIANCE to climb the TOWER.

Violation: Supported ROBOTS become ineligible for TOWER points for the remainder of the MATCH.

7.4.4 Opponent Interaction

Note, [G415](#), [G416](#), [G417](#) are mutually exclusive. A single ROBOT to ROBOT interaction which violates more than 1 of these rules results in the most punitive penalty, and only the most punitive penalty, being assessed.

G415 *Stay out of other ROBOTS. A ROBOT may not use a COMPONENT outside its ROBOT PERIMETER (except its BUMPERS) to initiate contact with:

- A. an opponent ROBOT inside the vertical projection of the opponent's ROBOT PERIMETER, or
- B. with the opponent's BUMPER backing or mounting.

Contact with an opponent in an opening of their BUMPERS or in the space above the BUMPER opening are exceptions to this rule.

Violation: MINOR FOUL.

For the purposes of this rule, "initiate contact" is a judgement call as to which ROBOT(s) in any interaction were responsible for the contact occurring based on direction of travel and relative speeds of both ROBOTS. Generally, for a ROBOT to have initiated contact, at minimum they have to be both moving towards the opponent ROBOT and have reasonable possibility to have avoided the contact.

In a collision, it's possible for both ROBOTS to initiate contact.

G416 *This isn't combat robotics. A ROBOT may not damage or functionally impair an opponent ROBOT in either of the following ways:

- A. deliberately.
- B. regardless of intent, by initiating contact, either directly or transitively via a SCORING ELEMENT CONTROLLED by the ROBOT:
 - a. inside the vertical projection of an opponent's ROBOT PERIMETER, or
 - b. with the opponent's BUMPER backing or mounting.

Contact between the ROBOT'S BUMPERS or COMPONENTS inside the ROBOT PERIMETER and COMPONENTS inside an opening of an opponent's BUMPERS or in the space above the BUMPER opening are exceptions to this rule.

Damage or functional impairment because of contact with a tipped-over opponent ROBOT, which is not perceived by a REFEREE to be deliberate, is not a violation of this rule.

Violation: MAJOR FOUL and YELLOW CARD, or if opponent ROBOT is unable to drive, then MAJOR FOUL and RED CARD.

FIRST Robotics Competition can be a full-contact competition and may include rigorous game play. While this rule aims to limit severe damage to ROBOTS, teams should design their ROBOTS to be robust.

Examples of violations of this rule include, but are not limited to:

- A. A ROBOT leaves an arm extended, spins around to change course, and unintentionally hits and damages a COMPONENT inside the ROBOT PERIMETER of a nearby opponent ROBOT.
- B. A ROBOT, in the process of trying to quickly reverse direction, tips up on a single pair of wheels, lands atop an opponent ROBOT, and damages a COMPONENT inside that opponent's ROBOT PERIMETER.
- C. A ROBOT high-speed rams and/or REPEATEDLY smashes an opponent ROBOT and causes damage. The REFEREE infers that the ROBOT was deliberately trying to damage the opponent's ROBOT.

Examples of functionally impairing another ROBOT include, but are not limited to:

- D. opening an opponent's relief valve such that the opponent's air pressure drops and
- E. powering off an opponent's ROBOT (this example also clearly results in a RED CARD because the ROBOT is no longer able to drive).

At the conclusion of the MATCH, the Head REFEREE may elect to visually inspect a ROBOT to confirm violations of this rule made during a MATCH and remove the violation if the damage cannot be verified.

For the purposes of this rule, "initiating contact" is a judgement call as to which ROBOT(s) in any interaction were responsible for the contact occurring based on direction of travel and relative speeds of both ROBOTS. Generally, for a ROBOT to have initiated contact, at minimum they have to be both moving towards the opponent ROBOT and have reasonable possibility to have avoided the contact.

In a collision, it's possible for both ROBOTS to initiate contact.

"Unable to drive" means that because of the incident, the DRIVER can no longer drive to a desired location in a reasonable time (generally). For example, if a ROBOT can only move in circles, or can only move extremely slowly, the ROBOT is considered unable to drive.

The exception in [G416-B](#) effectively means that ROBOTS with BUMPER gaps are at their own risk regarding damaging contact in these areas.

G417 **Don't tip or entangle.* A ROBOT may not deliberately, attach to, tip, or entangle with an opponent ROBOT.

Violation: MAJOR FOUL and YELLOW CARD, or if CONTINUOUS or opponent ROBOT is unable to drive, then MAJOR FOUL and RED CARD.

Examples of violations of this rule include, but are not limited to:

- A. using a wedge-like MECHANISM to tip over opponent ROBOTS,
- B. making BUMPER-to-BUMPER contact with an opponent ROBOT that is attempting to right itself after previously falling over and causing them to fall over again, and
- C. causing an opponent ROBOT to tip over by contacting the ROBOT after it starts to tip if, in the judgement of the REFEREE, that contact could have been avoided.

Tipping as an unintended consequence of normal ROBOT to ROBOT interaction, including single BUMPER to BUMPER hits that result in a ROBOT tipping, is not a violation of this rule.

"Unable to drive" means that because of the incident, the DRIVER can no longer drive to a desired location in a reasonable time (generally). For example, if a ROBOT can only move in circles, or can only move extremely slowly, the ROBOT is considered unable to drive.

G418 *There's a 3-count on PINS. A ROBOT may not PIN an opponent's ROBOT for more than 3 seconds. A ROBOT is PINNING if it is preventing the movement of an opponent ROBOT by contact, either direct or transitive (such as against a FIELD element). A PIN count ends once any of the following criteria below are met:

- A. the ROBOTS have separated by at least 72.0in (1.83m) from each other for more than 3 seconds,
- B. either ROBOT has moved 72.0in (1.83m) from where the PIN initiated for more than 3 seconds, or
- C. the PINNING ROBOT gets PINNED.

For criteria A, the PIN count pauses once ROBOTS are separated by 72.0in (1.83m) until either the PIN ends or the PINNING ROBOT moves back within 72.0in (1.83m), at which point the PIN count is resumed.

For criteria B, the PIN count pauses once either ROBOT has moved 72.0in (1.83m) from where the PIN initiated until the PIN ends or until both ROBOTS move back within 72.0in (1.83m), at which point the PIN count is resumed.

Violation: MINOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed.

A team's desired direction of travel is not a consideration when determining if a ROBOT is PINNED.

G419 *Don't collude with your partners to shut down major parts of game play. 2 or more ROBOTS that appear to a REFEREE to be working together may not isolate or close off any major element of MATCH play.

Violation: MAJOR FOUL, and for every 3 seconds in which the situation is not corrected, a MAJOR FOUL is assessed.

Examples of violations of this rule include, but are not limited to:

- A. shutting down access to all SCORING ELEMENTS,
- B. quarantining all opponents to a small area of the FIELD,
- C. preventing access to the opponent's TOWER,
- D. preventing traversal between field zones by blocking both TRENCHES, and
- E. preventing traversal between field zones by blocking both BUMPS.

Examples of standard gameplay that are not violations, include, but are not limited to:

- F. A single ROBOT blocking access to a particular area of the FIELD, and
- G. 2 ROBOTS independently collecting SCORING ELEMENTS in front of a BUMP or TRENCH at the same time.

G420 TOWER protection. A ROBOT may not contact, directly or transitively through a SCORING ELEMENT, an opponent ROBOT in contact with an opponent TOWER during the last 30 seconds of the MATCH regardless of who initiates contact.

Violation: MAJOR FOUL and if the opponent ROBOT is off the ground the opponent ROBOT is awarded LEVEL 3 TOWER points.

Transitive contact requires that each robot is contacting the same SCORING ELEMENT(S) simultaneously.

7.4.5 Human

G421 ***No wandering.** A DRIVE TEAM member must remain in their designated area as follows:

- A. DRIVERS and DRIVE COACHES may not contact anything outside their ALLIANCE AREA,
- B. a DRIVER must use the OPERATOR CONSOLE in the DRIVER STATION to which they are assigned, as indicated on the team sign,
- C. a HUMAN PLAYER may not contact anything outside their ALLIANCE AREA, and
- D. a TECHNICIAN may not contact anything outside their designated area.

Exceptions are granted as follows:

- E. for a HUMAN PLAYER partially outside the ALLIANCE AREA,
- F. in cases concerning safety, and
- G. for actions that are inadvertent, MOMENTARY, and inconsequential.

Violation: MINOR FOUL.

An intent of item B is to prevent unsafe situations where long tethers to OPERATOR CONSOLE devices increase tripping hazards as the operator moves about the ALLIANCE AREA. In the interest of avoiding nuisance penalties associated with a DRIVE TEAM member stepping outside of a prescribed area, we prefer to offer a general guideline as to what it means to use the OPERATOR CONSOLE in the ALLIANCE AREA. Provided the DRIVE TEAM member is within close proximity of their DRIVER STATION, there will be no repercussions. However, a DRIVE TEAM member located more than approximately half a DRIVER STATION width away from their own DRIVER STATION while using their OPERATOR CONSOLE is likely violating this rule.

G422 ***DRIVE COACHES and other teams: hands off the controls.** A ROBOT shall be operated only by the DRIVERS and/or HUMAN PLAYERS of that team. A DRIVE COACH activating their E-Stop or A-Stop is the exception to this rule.

Violation: MAJOR FOUL. RED CARD if greater-than-MOMENTARY.

Exceptions may be made before a MATCH for major conflicts, e.g. religious holidays, major testing, transportations issues, etc.

G423 ***DRIVE TEAMS, watch your reach.** A DRIVE TEAM member may not extend:

- A. into the CHUTE beyond the ALLIANCE-colored tape line while the CHUTE DOOR is open, or
- B. into the CORRAL beyond the ALLIANCE-colored tape line.

Violation: MINOR FOUL.

Teams should be careful when placing or collecting SCORING ELEMENTS and be aware of [G101](#).

- G424** ***Humans: use SCORING ELEMENTS as directed.** A DRIVE TEAM member may not deliberately use a SCORING ELEMENT in an attempt to ease or amplify a challenge associated with a FIELD element.

Violation: MAJOR FOUL.

An example of a violation of this rule is if a HUMAN PLAYER uses a FUEL to disrupt an opponent ROBOT attempting to climb on the TOWER.

- G425** ***SCORING ELEMENT delivery.** FUEL may only be introduced to the FIELD by a HUMAN PLAYER or DRIVER in the following ways:

- A. through the CHUTE,
- B. through the bottom opening in the OUTPOST, or
- C. thrown over the top of the ALLIANCE WALL from the OUTPOST AREA.

Violation: MAJOR FOUL.

- G426** ***DRIVE COACHES, SCORING ELEMENTS are off limits.** DRIVE COACHES may not touch SCORING ELEMENTS, unless for safety purposes.

Violation: MINOR FOUL.

- G427** **The OUTPOST has a storage limit.** Off-FIELD FUEL may only be stored in the CHUTE and the CORRAL. Excess FUEL, defined as the CHUTE & CORRAL being full, must immediately be entered onto the FIELD. HUMAN PLAYERS holding up to 2 FUEL or making a good-faith effort to immediately move or enter additional FUEL is an exception to this rule.

Violation: MINOR FOUL, and if CONTINUOUS, a MAJOR FOUL is assessed.

7.5 Post-MATCH

- G501** ***Leave promptly.** A DRIVE TEAM member may not cause significant or multiple delays to the start of a subsequent MATCH, scheduled break content, or other FIELD activities.

Violation: VERBAL WARNING. YELLOW CARD if subsequent violations at any point during the event.

